

Alternative Asymptotics and Fixed-Alternative Power Analysis ^{*}

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Abstract

When constructing Wald tests, consistency is the key property required for the variance estimator. This property ensures asymptotic validity of Wald tests and confidence intervals. Classical efficiency comparisons of hypothesis tests indicate all consistent variance estimators lead to equivalent Wald tests under local power approximations. This paper develops a simple asymptotic framework under fixed alternatives, which leads to new conclusions. In particular, we identify that variance estimation will have a first-order impact on the efficiency of Wald tests when size tends to zero with sample size while effect sizes are fixed. We apply this framework to several applications, including cluster-robust inference and quantile regression. In the case of cluster-robust inference, we provide for an asymptotic framework in which choosing the wrong level of clustering can lead to different power properties asymptotically. Numerical evidence suggests the results are applicable to moderate sample sizes and that the results can provide a useful approximation in standard ranges of size and power.

1 Introduction

Much of empirical work in economics follows a three step recipe: estimate the parameter of interest, estimate the asymptotic variance, then construct a test statistic or confidence interval to answer the research question. The first step is generally treated differently than the other two; while discussions on parameter estimation often focus on efficiency, the dialogue around variance estimation and testing typically focuses on robustness to misspecification. Ignoring any efficiency implications of variance estimation is at odds with the lived experience of empirical researchers, in that robust variance estimators are often considered “conservative.” In this paper we provide a first-order asymptotic theory which characterises scenarios in which variance estimation affects the asymptotic power of tests. The resulting asymptotic theory provides a theoretical foundation for several common “folk” theorems in applied work. This paper also clarifies the setting in which variance estimation is important for efficiency considerations: when the alternative is large and size is small.

The choice of variance estimator is an ever present decision in applied work. There is a menu of available robust, consistent variance estimators in standard statistical packages. Researchers with grouped data must determine whether to compute cluster-robust standard errors, and what level to cluster at. In the case of quantile regression, researchers choose a kernel density estimator to use. In likelihood settings under correct

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specification, the Fisher information matrix can be estimated using the outer product of the score or the second derivative of the log-likelihood. Robust variance estimators in time series involve choosing a kernel and truncation point. We will not consider all these examples here, however we provide a framework that is suited to studying the effect of variance estimation in many of these contexts.

This paper makes three key contributions to the econometric literature on hypothesis testing. First, we clarify the settings in which variance estimation can impact the power of tests. When the alternative is large relative to sample size and size tends to 0, test statistics using different variance estimators will have different first-order asymptotic properties. The limiting power along the sequence of tests depends not only on consistency of the variance estimator, but also the variance of the *variance estimator* and possible covariance between the variance estimator and point estimator.

Second, we present some new insights on sample size calculation and connections to Bahadur asymptotic relative efficiency. In the Bahadur setting, size tends to zero as sample size increases, and power converges to a constant. We point out that this setting in fact does not lend itself well to sample size calculations as the limiting power drops out of the analysis. Our numerical evidence suggests that our approach is competitive with local power when we aim to find the minimum sample size to achieve a certain power under a particular alternative.

Last, we apply this approach to several applications, including smooth generalized method of moments (GMM) problems, linear models with possible cluster dependence, and quantile regression.

The theory developed in this paper takes a different approach compared with the traditional local-asymptotic theory of Pitman (1949), Engle (1984), Newey and McFadden (1994), van der Vaart (1998), and Le Cam (1986). That work finds that a broad class of tests statistics have the same limiting distribution under local-alternatives. Our analysis is non-local, which leads to these equivalencies no longer holding in general. Under fixed alternatives, we can view a re-scaled version of the tests statistic as an estimator of a non-centrality parameter: the ratio of the effect size to the variance. With this perspective in mind, the asymptotic behaviour of the test statistic depends on both the estimation of the numerator (depending on the parameter estimator) and the denominator (depending on the variance estimator). In the case of Wald tests, local equivalence holds whenever the same parameter estimator is used in two different tests, even if different consistent variance estimators are used. This equivalence no longer holds in our asymptotic theory when different estimators of the asymptotic variance are used. It turns out that variance estimation has an impact on asymptotic power in testing regimes where size converges to 0 and power converges to some fixed constant. These asymptotics have a similar flavour to the asymptotic experiment proposed in Bahadur (1967). A benefit of our approach is that no large-deviation results are necessary. In a similar fashion, our approach is also more broadly applicable than the measure proposed in Hodges and Lehmann (1956), where size converges to a constant and power converges to 1, an approach which also requires large-deviation theory.

We are not the first to use asymptotics to assess the behaviour of different variance estimators. Hall (2000) points out that tests constructed with variance estimators which are consistent under the null and alternative hypothesis are more efficient than those using a variance estimator which is only efficient under the null. Sun, Phillips, and Jin (2008) compares fixed-bandwidth to small bandwidth asymptotics in constructing standard errors for time-series regressions. They note that traditional optimal choices of the bandwidth are chosen based on estimation criteria, rather than testing criteria, and develop a theory where the power and size of tests depend explicitly on bandwidth choice in a variety of asymptotic frameworks. Iacone, Leybourne, and Taylor (2013) uses fixed-b asymptotics to choose between tuning parameters for variance estimators and test

statistics when testing for breaks in a linear trend. This paper links the estimation of the variance to the power of tests along a different sequence of tests where size tends to zero. In Kato (2012), the asymptotic distribution is derived for the kernel density estimator for the asymptotic variance in the quantile regression setting of Koenker and Bassett (1978), for the particular choice of a uniform kernel. We extend the results in Kato (2012), and link the form of the asymptotic bias and variance to power calculations.

There has also been other work in econometrics on ways of characterising the behaviour of tests under the alternative that also differ from the local asymptotic normality approach of Le Cam (1986). Kim and Perron (2009) propose using an approximate version of the Bahadur (1967) asymptotic relative efficiency measure when comparing tests for structural breaks in time series. Canay and Otsu (2012) used Hodges-Lehmann asymptotic relative efficiency to assess the efficiency of GMM and generalized empirical likelihood tests of moments conditions. A benefit of our approach is broad applicability to testing problems most frequently encountered in empirical work, while maintaining an exact asymptotic comparison. The approach in this paper is perhaps most similar to Bahadur (1960a), Bahadur (1960b), and Bahadur (1967). In that setting, the rate at which p-values tend to zero under an alternative are connected to the rate at which size can be sent to zero and a fixed non-trivial power can be attained asymptotically. As discussed in Groeneboom and Oosterhoff (1981), this approach presents some challenge for a practical approach to asymptotical relative efficiency in that the limiting non-trivial power does not show up in the analysis of Bahadur asymptotic relative efficiency. We do not make *general* claims on ranking tests using our power analysis, as there is no guarantee in the general case that size tends to zero at the same rate across tests. On the other hand, our analysis provides for non-trivial power calculations when size is small and the alternative is large relative to sample size. In Section 4, in the context of the t-test with homogenous clusters and Gaussian observations, we do derive a result mapping the power comparison we consider to an efficiency result.

We demonstrate the broad applicability of our approach by considering several important applications of the theory. We derive an asymptotic approximation of Wald test statistics for general use in GMM problems. The main result is quite general, allowing for cluster-dependence and non-smooth moment conditions. The first specific application we provide is to cluster-robust inference in the linear model. The general framework for cluster-robust inference we adopt is that in Hansen and Lee (2019). An alternative approach, the design-based approach, is discussed in Abadie, Athey, Imbens, and Wooldridge (2020). Popularised in Bertrand, Duflo, and Mullainathan (2004), some recent work in econometrics has focused on the choice of cluster level. In Cameron and Miller (2015) it is argued that the coarsest cluster level should always be used. Abadie, Athey, Imbens, and Wooldridge (2023) presents a design-based approach to choosing the appropriate cluster level, along with some finite-sample results. MacKinnon, Nielsen, and Webb (2023b) provide a sequential testing procedure to detect the correct clustering level, within the model-based framework we adopt. We show that there is an unambiguous loss of efficiency when independent observations are included in the same cluster. Our results imply a method for researchers to conduct power analysis to see if the efficiency loss in their case is severe, or if there is little to be lost from the added robustness.

Our second specific application is to quantile regression. We focus on the linear conditional quantile regression model of Koenker and Bassett (1978). In this case, classic approaches to variance estimation involve estimators of the conditional density of the error term. We focus on the kernel density estimator of Powell (1991). The default choices in the `quantreg` package in R and the `qreg` function in Stata are the Gaussian and Epanechnikov kernel, respectively. We show that the choice of kernel and bandwidth do have an impact on the first-order behaviour of the test statistics in our setting.

Our distributional theory extends results in Bentkus, Jing, Shao, and Zhou (2007), Omey and van

Gulck (2009), and Shao and Zhang (2009), where one-sample t-statistics and similar types of statistics are considered. We extend the basic theory to GMM problems under sampling with cluster-dependence, including non-smooth problems such as quantile regression. In Bentkus et al. (2007) these asymptotic distributions are used to motivate asymptotic power functions. We use this type of calculation to motivate our own relative efficiency comparison.

The rest of the paper proceeds as follows: we start by introducing the principles of our analysis in Section 2, in the context of a simple testing problem: hypothesis testing for means. In Section 3, we provide a treatment of the distribution of Wald statistics in GMM settings, under fixed alternatives. A discussion on using the framework for power analysis is discussed in Section 4 and some simulations are provided in Section 5 to show the efficacy of the methods here in making finite-sample predictions. A summary of our results is discussed in Section 6. Throughout, Φ is used to denote the standard normal cumulative distribution function and $\Rightarrow$ is used to denote convergence in distribution. $q_{1-\alpha,\nu}$ and $z_{1-\beta}$ are used to denote the $1-\alpha$ quantile of a χ^2_ν random variable and the $1-\beta$ quantile of a standard normal random variable respectively. We use the standard stochastic order symbols O_P and o_P to mean bounded in probability and converges to zero in probability respectively.

2 An Illustrative Example: the Sample Mean

We begin by considering a simple testing problem: a two-sided hypothesis test for the sample mean. To illustrate the basic approach, we compare the classic Wald test statistic with a cluster-robust version. Under sequences of local-alternatives, these test statistics have the same asymptotic properties, and therefore the same asymptotic power function. Hence, it is not possible to use asymptotic methods to compare such tests. To overcome this, we compare the asymptotic distributions under fixed alternatives. In this asymptotic setting we find that the asymptotic distributions differ for the two test statistics. This leads to a natural relative efficiency comparison, in which we find that when the observations are independent (i.e. both test statistics have correct asymptotic size) there is an asymptotic power loss associated with using the cluster-robust test statistic.

2.1 Cluster-robust inference

Consider a sample $\{X_{ig}\}$, where i denotes observation i in group g . There are G equal-sized groups, each containing n_g observations, for a total of $Gn_g = n$ observations. For the purposes of this example, we consider the case that n_g is fixed, so that $n/G = n_g$ for all sample sizes.¹ A concerned researcher suggests that we should use cluster-robust methods since the data were grouped when collected, however we know that the observations are independent and identically distributed. For all i, g , we have that $\mathbb{E} X_{ig} = \mu$ and $\text{Var}(X_{ig}) = \sigma^2$. We would like to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$. We construct Wald tests based on the sample mean:

$$\bar{X}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} X_{ig}.$$

¹We can also accommodate unbalanced designs with growing cluster sizes; this type of result is also covered in Section 3.

We compare this test statistic, the classic Wald test statistic, to a cluster-robust version.² The classic Wald test statistic is

$$W_h = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_h^2}, \quad \hat{\sigma}_h^2 = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} (X_{ig} - \bar{X}_n)^2. \quad (2.1)$$

The cluster-robust Wald statistic is

$$W_c = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_c^2}, \quad \hat{\sigma}_c^2 = \frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} (X_{ig} - \bar{X}_n) \right)^2. \quad (2.2)$$

where since we will have $G \rightarrow \infty$ the estimator is still consistent for σ^2 .

Under the null hypothesis, and without any cluster dependence, we have that each statistic has the same asymptotic distribution,

$$\begin{aligned} W_h &\Rightarrow \chi_1^2 \\ W_c &\Rightarrow \chi_1^2 \end{aligned}$$

which follows from a basic application of Slutsky's theorem. The numerator of each test statistic, divided by σ^2 , is asymptotically χ_1^2 . In the denominator of each test statistic is a consistent estimator of the variance. This first-order analysis effectively considers the properties of the test statistics under the simplifying assumption that the variance estimator is equal to its probability limit.

The same logic holds in the case of a sequence of local alternatives. Consider $\mu_n = \mu_0 + \delta/\sqrt{n}$. In this case, Slutsky's theorem applies again: the only change is that the numerator of the test statistic is no longer correctly centered, therefore the limiting distribution for each test-statistic is $\chi_1^2(\delta^2/\sigma^2)$. Local power analysis cannot distinguish between a test statistic constructed using $\hat{\sigma}_h^2$ and one constructed using $\hat{\sigma}_c^2$. This asymptotic equivalence leads to the common conclusion among econometricians that the choice of variance estimator does affect the power of the test.

Now consider the fixed alternative $\mu = \Delta + \mu_0$. We are testing $H_0 : \mathbb{E} X_{ig} = \mu_0$. Define the non-centrality parameter:

$$\xi := \frac{\Delta}{\sigma}.$$

For $a \in \{h, c\}$, expansions of the test statistics under a fixed alternative gives

$$\frac{1}{n} W_a = \frac{(\bar{X}_n - \mu)^2}{\hat{\sigma}_a^2} + \frac{2\Delta(\bar{X}_n - \mu)}{\hat{\sigma}_a^2} + \frac{\Delta^2}{\hat{\sigma}_a^2}. \quad (2.3)$$

The first two terms converge in probability to 0, and the last term converges to ξ^2 in each case, and thus we can say that $\frac{1}{n} W_a \xrightarrow{P} \xi^2$. Thus, one way of viewing the test statistic under a fixed alternative is as a scaled estimator of the non-centrality parameter ξ^2 . In (2.3), the first term on the right-hand side is asymptotically negligible relative to the other two terms. If $\mathbb{E} X_{ig}^4 < \infty$, then the variance estimators are also asymptotically normal. For the homoskedastic variance estimator, this is a straightforward application of the central limit

²For simplicity, we do not include any degrees-of-freedom correction, whether for the variance estimator or number of clusters. Since we use the large- G asymptotics of Hansen and Lee (2019), these degrees of freedom corrections disappear in the limit.

theorem:

$$\begin{aligned}
\sqrt{n}(\hat{\sigma}_h^2 - \sigma^2) &= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} ((X_{ig} - \bar{X}_n)^2 - \sigma^2) \\
&= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} ((X_{ig} - \mu)^2 - \sigma^2) - \sqrt{n}(\bar{X}_n - \mu)^2 \\
&\Rightarrow \mathcal{N}(0, \sigma^4(\kappa - 1))
\end{aligned}$$

where κ is the kurtosis of X_{ig} . For the cluster-robust variance estimator, a computationally convenient way to understand the asymptotic behaviour is to consider the cluster-averages $\tilde{X}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} X_{ig}$ as the observations rather than the X_{ig} . Using the relationships between central moments and cumulants, as well as the independence of the X_{ig} , we have that

$$\begin{aligned}
\sqrt{n}(\hat{\sigma}_c^2 - \sigma^2) &= \sqrt{n} \left(\frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} (X_{ig} - \bar{X}_n) \right)^2 - \sigma^2 \right) \\
&= \sqrt{n} n_g \left(\frac{1}{G} \sum_{g=1}^G \left((\tilde{X}_g - \mu)^2 - \frac{\sigma^2}{n_g} \right) - (\bar{X}_n - \mu)^2 \right) \\
&= n_g^{3/2} \frac{1}{\sqrt{G}} \sum_{g=1}^G \left((\tilde{X}_g - \mu)^2 - \frac{\sigma^2}{n_g} \right) - n_g \sqrt{n} (\bar{X}_n - \mu)^2 \\
&\Rightarrow \mathcal{N}(0, \sigma^4(\kappa - 1) + 2\sigma^4(n_g - 1)). \tag{2.4}
\end{aligned}$$

The variance of the cluster-robust variance estimator is inflated relative to the homoskedastic version by a factor of $2\sigma^4(n_g - 1)$. If we consider the standardised difference between the estimators $\sqrt{n}(\hat{\sigma}_c^2 - \hat{\sigma}_h^2)$, we obtain that

$$\begin{aligned}
\sqrt{n}(\hat{\sigma}_c^2 - \hat{\sigma}_h^2) &= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i \neq j} (X_{ig} - \mu)(X_{jg} - \mu) - \sqrt{n}(n_g - 1)(\bar{X}_n - \mu)^2 \\
&= \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i \neq j} (X_{ig} - \mu)(X_{jg} - \mu) + o_P(1).
\end{aligned}$$

When the observations X_{ig} are i.i.d., these terms are mean-zero, and thus do not alter the conclusion that $\hat{\sigma}_c^2 - \hat{\sigma}_h^2 \xrightarrow{P} 0$, however they do contribute to the variance:

$$\begin{aligned}
\text{Var} \left(\frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i \neq j} (X_{ig} - \mu)(X_{jg} - \mu) \right) &= \frac{1}{G} \sum_{g=1}^G \frac{1}{n_g} \text{Var} \left(\sum_{i \neq j} (X_{ig} - \mu)(X_{jg} - \mu) \right) \\
&= \frac{1}{n_g} \text{Var} \left(\sum_{i \neq j} (X_{ig} - \mu)(X_{jg} - \mu) \right) \\
&= \frac{1}{n_g} \mathbb{E} \left(\sum_{i \neq j} (X_{ig} - \mu)(X_{jg} - \mu) \right)^2
\end{aligned}$$

The sum has $n_g(n_g - 1)$ terms of the form $(X_{ig} - \mu)(X_{jg} - \mu)$. Expanding the square, we end up with

$n_g(n_g - 1)$ terms $(X_{ig} - \mu)^2(X_{jg} - \mu)^2$ from squaring each entry, as well as an additional identical $n_g(n_g - 1)$ terms from the product of $(X_{ig} - \mu)(X_{jg} - \mu)$ with $(X_{jg} - \mu)(X_{ig} - \mu)$. All other products that come out of the sum are mean zero: $\mathbb{E}((X_{ig} - \mu)(X_{jg} - \mu)(X_{lg} - \mu)(X_{mg} - \mu)) = 0$ unless $i = l$ and $j = m$ or $i = m$ and $j = l$. Thus, the variance is made up of $2n_g(n_g - 1)$ terms, each equal to $\mathbb{E}(X_{ig} - \mu)^2(X_{jg} - \mu)^2 = \sigma^4$. In this way, we can understand the asymptotic variance in (2.4) as a sum of two variances. $\sigma^4(\kappa - 1)$ is the asymptotic variance of $\hat{\sigma}_h^2$, and is related to the fourth-moments of the X_{ig} . The $2\sigma^4(n_g - 1)$ term is the variance of the additional terms that must be included when achieving robustness with respect to cluster dependence.

Now, consider again the expansion (2.3). For the distribution of $\sqrt{n}(\frac{1}{n}W_a - \xi^2)$, we need the joint limit distribution of $\bar{X}_n$ and $\hat{\sigma}_a^2$. Defining γ as the skewness of X_{ig} and applying the delta-method to (2.3), we have

$$\sqrt{n} \left(\frac{1}{n}W_a - \xi^2 \right) \Rightarrow \mathcal{N}(0, \Sigma_a) \quad (2.5)$$

where

$$\Sigma_h = \xi^2((\kappa - 1)\xi^2 - 4\gamma\xi + 4) \quad (2.6)$$

$$\Sigma_c = \Sigma_h + 2(n_g - 1)\xi^4. \quad (2.7)$$

In the application of the delta-method to (2.3) we recognise that the first term is $O_P(1/n)$: it is the Wald statistic to test the null hypothesis $H_0 : \mathbb{E}X_{ig} = \mu$. Thus, it is the second two terms that are important for the asymptotic distribution of $\frac{1}{n}W_a$. We then have

$$\frac{1}{n}W_a - \xi^2 = \frac{2\Delta(\bar{X}_n - \mu)}{\hat{\sigma}_a^2} + \frac{\Delta^2}{\hat{\sigma}_a^2} - \frac{\Delta^2}{\sigma^2} + O_P(1/n) \quad (2.8)$$

$$= \frac{2\Delta}{\sigma^2}(\bar{X}_n - \mu) - \frac{\Delta^2}{\sigma^4}(\hat{\sigma}_a^2 - \sigma^2) + O_P(1/n) + o_P(|\hat{\sigma}_a^2 - \sigma^2|) \quad (2.9)$$

This stochastic expansion is valid whenever $\sigma^2 \geq c > 0$ for some c , and will be useful to recall later when looking at the general case where the convergence rate for $\hat{\sigma}_a^2$ and $\bar{X}_n$ might be different. We can see from (2.9) that characterizing the asymptotic distribution of the Wald test statistics under a fixed alternative is equivalent to characterizing the asymptotic distribution of a linear combination of the estimator from the numerator, $\bar{X}_n$, and the estimator of the variance, $\hat{\sigma}_a^2$. Since we have already shown that the variance estimators in this sample have different first-order behaviour, the limit distribution of W_a (properly centered and normalized) will vary depending on which variance estimator is used.

The computationally convenient expansion of $\hat{\sigma}_c^2$ in terms of the cluster averages $\tilde{X}_g$ gives one interpretation of the difference between the two test statistics: the cluster-robust variance estimator is effectively using only G observations for estimating the variance, rather than n , thereby inflating the variance by a factor of approximately $n/G = n_g$. The natural interpretation based on looking at the difference $\hat{\sigma}_c^2 - \hat{\sigma}_h^2$ is that the robustness provided by $\hat{\sigma}_c^2$ leads to additional terms in the variance estimator. These terms are mean-zero, and therefore drop out while applying Slutsky's theorem under local alternatives, however they contribute to the variance of the test statistic under an alternative which is a fixed distance from the null.

We now connect the asymptotic distribution of the test statistics to power. Local alternatives give a (local) asymptotic approximation to power: under the sequence of local alternatives $\mu_n = \mu_0 + \delta/\sqrt{n}$, we have that

$$P(W_a > q_{1-\alpha,1}) \rightarrow 1 - F_{\chi_1^2(\delta^2/\sigma^2)}(q_{1-\alpha,1}), \quad a \in \{h, c\}. \quad (2.10)$$

The limiting power function is that of a non-central chi-square distribution which is invariant to the choice of variance estimator. The asymptotic power comparisons under local alternatives do not distinguish between Wald tests where different consistent variance estimators are used.

One implication of (2.5), (2.6), and (2.7) is that under fixed alternatives the test statistics have different asymptotic distributions. This implies that we can design a sequence of tests which leads to non-degenerate power under the fixed alternative $\mathbb{E} X_{ig} = \mu = \mu_0 + \Delta$. Let C_n^a denote a sequence of critical values for each test statistic. Using the asymptotic distribution in (2.5), we have that

$$\begin{aligned} P(W_a > C_n^a) &= P\left(\left(\frac{1}{n}W_a - \xi^2\right) > \frac{C_n^a}{n} - \xi^2\right) \\ &= P\left(\Sigma_a^{-1/2}\sqrt{n}\left(\frac{1}{n}W_a - \xi^2\right) > \frac{C_n^a}{\sqrt{n}\Sigma_a} - \xi^2\sqrt{\frac{n}{\Sigma_a}}\right). \end{aligned} \quad (2.11)$$

This suggests a choice for C_n^a . If we choose

$$C_n^a = n\left(\xi^2 - t\sqrt{\frac{\Sigma_a}{n}}\right) \quad (2.12)$$

then we have that under the alternative $\mathbb{E} X_{ig} = \mu_0 + \Delta$,

$$P(W_a > C_n^a) = P\left(\Sigma_a^{-1/2}\sqrt{n}\left(\frac{1}{n}W_a - \xi^2\right) > -t\right) \rightarrow \Phi(t) \quad (2.13)$$

Part of the appeal of local power analysis is that asymptotically power is in $(0, 1)$. Under a fixed alternative, this sequence of critical values will lead to such nondegenerate power in the limit. If we choose $t = z_{1-\beta}$, then $P(W_a > C_n^a) \rightarrow 1 - \beta$, noting that the sequence of critical values will be different for each test statistic to achieve the same power. The chosen sequence C_n^a is local to the (scaled) non-centrality parameter $n\xi^2$.

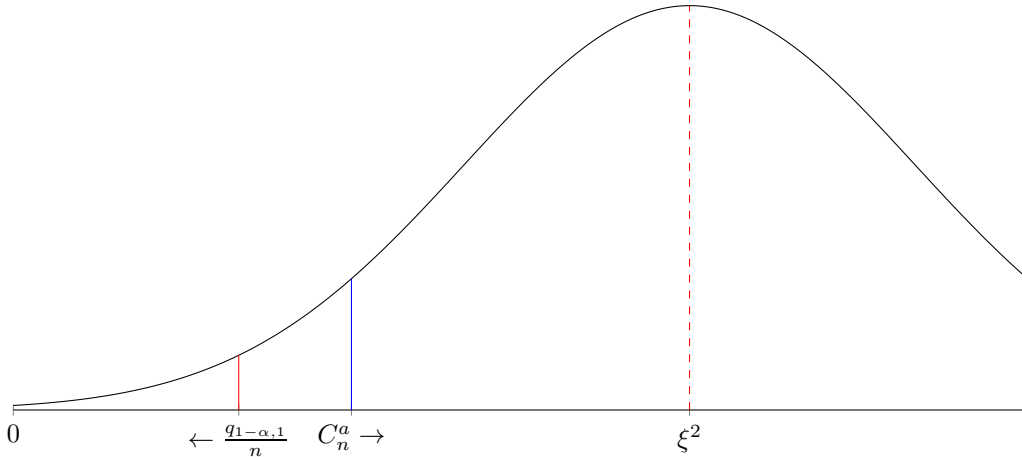


Figure 1: The distribution of $\frac{1}{n}W_a$ will concentrate around the non-centrality parameter ξ^2 , while the critical value for the test $\frac{1}{n}q_{1-\alpha,1}$ converges to 0. The chosen sequence C_n^a will converge to the non-centrality parameter at the correct rate so that the power of this sequence of tests is non-degenerate asymptotically.

In Figure 1, we lay out the relationship between the non-centrality parameter, C_n^a , and $\frac{1}{n}q_{1-\alpha,1}$. We want to gain insight into asymptotic behaviour of the exact power in (2.11), and therefore choose C_n^a converging

to ξ^2 . Note that any sequence C_n such that $\sqrt{n}(C_n - n\xi^2)$ tends to a constant will have this property, and the choice given is for convenience. A similar asymptotic power approximation was proposed in Bentkus et al. (2007) in the case of the 1-sample t-test. In that paper they were primarily concerned with variations in behaviour of the t-test under a fixed alternative when the observations had possibly fewer than 4 finite moments.

For our two test statistics we have thus considered, the considerations on power lead to

$$C_n^h = n \left(\xi^2 - z_{1-\beta} \sqrt{\frac{\xi^2 ((\kappa - 1)\xi^2 - 4\gamma\xi + 4)}{n}} \right)$$

$$C_n^c = n \left(\xi^2 - z_{1-\beta} \sqrt{\frac{\xi^2 ((\kappa - 1)\xi^2 - 4\gamma\xi + 4 + 2(n_g - 1)\xi^2)}{n}} \right).$$

Along these sequences, we have that

$$P(W_n^h > C_n^h) \rightarrow 1 - \beta$$

$$P(W_n^c > C_n^c) \rightarrow 1 - \beta.$$

Because the two test statistics have different asymptotic distributions under a fixed-alternative, difference sequences of critical values are required to achieve the same power. Thus, this analysis has lead to a sequence of testing problems which is sensitive to the choice of variance estimator. Since we have not discussed Type-I errors along this sequence, it is not clear how this should imply a preference-ordering between the two test statistics. The main conclusion to be drawn at this stage is that a sufficient condition for the first-order properties of a sequence of tests to depend on the choice of variance estimator is that size converges to 0.

The given choice of critical values implies that the size of the sequence of tests converges to 0. This feature is reminiscent of Bahadur asymptotic relative efficiency (Bahadur (1960a), Bahadur (1960b), and Bahadur (1967); see also Serfling (2009)). Applying that setup to our simple example, the comparison is based on the rate at which the p-values converge to 0 under a fixed alternative. Let $F_{0,n}^a$ be the CDF of the test statistics W_n^a under the null hypothesis. The p-values are then $1 - F_{0,n}^a(W_n^a)$. The Bahadur slope of the test $c(\Delta)$ is the rate at which the p-values converge to 0,

$$\lim_{n \rightarrow \infty} -\frac{2 \log(1 - F_{0,n}^a(W_n^a))}{n} = c_a(\Delta).$$

In that setup, $c_a(\Delta) > 0$ when W_n^a satisfies a large-deviation property under the null hypothesis. If such a property holds (as is the case, for example, when the X_{ig} are normally distributed) then we can compare the functions $c_a(\Delta)$ to compare the test statistics, with larger values being better as they reflect a faster convergence rate of the p-values to 0. After covering the general case of asymptotic power for Wald test statistics under fixed-alternatives, we return to how our approach augments the Bahadur approach for comparing test statistics in Section 4.

3 Fixed-Alternative Asymptotics

The previous section motivates the following derivation of the asymptotic distribution of Wald test statistics under fixed alternatives. In this section we introduce the general setup for deriving the asymptotic distribution of Wald test statistics under fixed alternatives for asymptotically linear estimators. We first give

sufficient conditions for when the test statistics are asymptotically normal under a fixed alternative, and then use this first-order approximation to motivate an alternative approach to power calculations.

3.1 Test Statistics under Fixed-Alternatives

We focus our attention on parameters which are identified by estimating equations:

$$\mathbb{E} \psi(X_{ig}, \theta) = 0 \quad (3.1)$$

where $X_{ig} \in \mathcal{X}$, $\theta \in \Theta \subset \mathbb{R}^p$, and $\psi : \mathcal{X} \times \Theta \rightarrow \mathbb{R}^k$, with $k \geq p$. Our focus will be on tests of a linear hypothesis. Define the parameter $\vartheta := \ell' \theta$ for a fixed vector ℓ . Tests of $H_0 : \vartheta = \vartheta_0$ against $H_1 : \vartheta \neq \vartheta_0$ are commonly performed in applied work using the Wald statistic:

$$W_n := n \frac{(\ell' \hat{\theta} - \vartheta_0)^2}{\ell' \hat{V}_n \ell} \quad (3.2)$$

where $\hat{V}_n$ is a suitable variance estimator of a constant matrix V_n such that $V_n^{-1/2} \sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}(0, I_p)$. Under standard regularity conditions, W_n is asymptotically χ_p^2 .

When $\vartheta \neq \vartheta_0$, W_n will diverge off to ∞ , in the sense that for all $C > 0$, $P(W_n > C) \rightarrow 1$. This is clearly not useful for characterising the behaviour of W_n in large samples, and therefore we will consider suitable centring and (re)scaling. Our first observation towards this goal is that under the same conditions providing for W_n to be asymptotically χ_p^2 , for $\Delta = \ell' \theta - \vartheta_0$, we have that:

$$\frac{(\ell' \hat{\theta} - \vartheta_0)^2}{\ell' \hat{V}_n \ell} - \frac{\Delta^2}{\ell' V_n \ell} \xrightarrow{P} 0. \quad (3.3)$$

Under a fixed alternative we can view $\frac{1}{n} W_n$ as a consistent estimator of the noncentrality parameter $\xi_n^2 := \frac{\Delta^2}{\ell' V_n \ell}$. In the GMM setting we consider here, $\frac{1}{n} W_n$ is a function of three random components: $\hat{\theta}$, $\hat{\Omega}_n$, and $\hat{Q}_n$, with $\hat{V}_n = (\hat{Q}_n' \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1}$ taking the usual sandwich form. Under regularity conditions on $\hat{Q}_n$ and $\hat{\Omega}_n$ and their probability limits, we will be able to write the test statistic as an asymptotically linear functional of three terms under a fixed alternative,

$$\frac{1}{n} W_n - \xi_n^2 \approx p'_{1n}(\hat{\theta} - \theta) + \text{tr}(P_{1n}(\hat{Q}_n - Q_n)) + \text{tr}(P_{2n}(\hat{\Omega}_n - \Omega_n)). \quad (3.4)$$

If we can characterise the behaviour of these three terms, we can then derive the normalising sequence Ξ_n and possible bias term B_n such that

$$\Xi_n^{-1/2} \left(\frac{1}{n} W_n - \xi_n^2 - B_n \right) \Rightarrow \mathcal{N}(0, 1). \quad (3.5)$$

We can then conclude that for critical values of the form

$$C_n = n(\xi_n^2 + B_n) - n z_{1-\beta} \sqrt{\Xi_n} \quad (3.6)$$

that $P(W_n > C_n) \rightarrow 1 - \beta$. In the following subsections we will give explicit expressions for p_{1n} , P_{1n} , P_{2n} , B_n , and Ξ_n in the case that the Wald statistic is formed using a plug-in estimator of V_n using the efficient GMM estimator $\hat{\theta}$. In the example of the t-test, $B_n = 0$ and Ξ_n is of the order $1/\sqrt{n}$, but we

will discuss a few examples where the more general case is of interest. Other estimators for $\hat{V}_n$ are also of interest theoretically, but are less common in practise. One exception is an estimator using $\tilde{\theta}_0$, an estimate of θ under the restriction $\ell'\theta = \vartheta_0$. We do not consider this case in general, however we will discuss the particular comparison of the score test for a normal mean below.

3.2 Assumptions

In this section we lay out our high-level conditions which allow us to derive the asymptotic distribution of Wald test statistics under fixed alternatives. As cluster-robust inference is one of our main applications, the first assumption places restrictions on the sizes of the clusters.

Assumption 1. (*Cluster dependence*) Denote X_{ig} as observation i in cluster g . Observations are independent across clusters, and we allow for arbitrary cluster dependence within cluster g , where there are G clusters. If cluster g has n_g observations, the overall sample size is $n = \sum_{g=1}^G n_g$. We require $G \rightarrow \infty$ as $n \rightarrow \infty$, and for some $r \in [2, \infty)$ and $C < \infty$, we have that

$$\frac{\left(\sum_{g=1}^G n_g^{2r}\right)^{2/r}}{n} \leq C, \quad \lim_{n \rightarrow \infty} \max_{g \in \{1, \dots, G\}} \left\{ \frac{n_g^4}{n} \right\} = 0. \quad (3.7)$$

This assumption is from Hansen and Lee (2019) and ensures that clusters cannot be too heterogeneous in size or grow too quickly with n for a central limit theorem to hold. The assumption is also similar to Assumption 3 in Djogbenou, MacKinnon, and Nielsen (2019). Here we use the version in Assumption 3 from Hansen and Lee (2019) paper, as we will be characterising the asymptotic behaviour of estimators of second moments. In this case, the estimator of the variance cannot be written as a sum over i , but is rather written as a sum over the clusters g , which requires the second part of (3.7) to have n_g^4 instead of n_g^2 . The constant r will be connected to the number of finite moments required below. Our conditions will imply that $\hat{\theta}$, $\hat{Q}_n$, and $\hat{\Omega}_n$ are asymptotically linear, and the influence function of each estimator will be required to have at least r moments, which at a minimum will require that $\mathbb{E} \|\psi(X_{ig}, \theta)\|^4 < \infty$ so that the variance estimator is asymptotically normal; if the clusters are particularly large or heterogenous in size, this will lead to stronger restrictions on the moments in the assumptions below. In several of our examples below, it is of sufficient interest to set $n_g = 1$ and $G = n$ and this assumptions is trivially satisfied. The assumption is also satisfied whenever n_g is fixed for all sample sizes and n is sufficiently large. It also applies when the clusters are growing, e.g. $n_g = n^\alpha$ for some $\alpha < (r-2)/2(r-1)$. For more discussion of the restrictions Assumption 1 places on the cluster sizes, see Hansen and Lee (2019) and Djogbenou et al. (2019).

The next assumption confirms the definition of θ , $\hat{\theta}$, and provides for W_n leading to valid inference when $\theta = \theta_0$. Before doing this we will define Q_n and Ω_n as the standard components of the asymptotic variance of $\hat{\theta}$ estimated using efficient GMM. Let $\dot{\psi}_{ig}(\theta) = \frac{\partial}{\partial \theta} \mathbb{E} \psi(X_{ig}, \theta)$ and $\tilde{\psi}_g(X_g, \theta) := \sum_{i=1}^{n_g} \psi_{ig}(X_{ig}, \theta)$. Then, we define

$$Q_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \dot{\psi}_{ig}(\theta) \quad (3.8)$$

$$\Omega_n := \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' \quad (3.9)$$

We will use plug-in estimators of Q_n and Ω_n , so that

$$\hat{Q}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}(X_{ig}, \hat{\theta}) \quad (3.10)$$

$$\hat{\Omega}_n := \frac{1}{n} \sum_{g=1}^G \hat{\psi}(X_g, \hat{\theta}) \hat{\psi}(X_g, \hat{\theta})' \quad (3.11)$$

for some function $\hat{\psi}$.

Assumption 2. We assume that there exists a unique $\theta \in \Theta \subset R^p$ such that $\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \psi(X_{ig}, \theta) = 0$. We further assume that $\mathbb{E} \psi(X_{ig}, b)$, $\mathbb{E} \hat{\psi}_{lm}(X_{ig}, b)$ and $\mathbb{E} \psi_l(X_{ig}, b) \psi_m(X_{ig}, b)$ are continuously differentiable in b for all indices l, m , for all b in an open neighbourhood N_θ which contains θ . We will denote $\frac{\partial}{\partial b} \mathbb{E} \psi(X_{ig}, b) =: \dot{\psi}_{ig}(b)$. We assume that

$$\lim_{M \rightarrow \infty} \sup_{i,g} \sup_{\theta \in N_\theta} \mathbb{E} \|\psi(X_{ig}, \theta)\|^{2r} \mathbf{1}_{[\|\psi(X_{ig}, \theta)\| > M]} = 0 \quad (3.12)$$

$$\lim_{M \rightarrow \infty} \sup_{i,g} \sup_{\theta \in N_\theta} \mathbb{E} \|\hat{\psi}(X_{ig}, \theta)\|^r \mathbf{1}_{[\|\hat{\psi}(X_{ig}, \theta)\| > M]} = 0 \quad (3.13)$$

where r satisfies Assumption 1. The matrices Q_n and Ω_n are full-rank, and the minimum eigenvalue $\lambda_{\min}(\Omega_n)$ satisfies

$$\lambda_{\min}(\Omega_n) \geq c > 0$$

for some c . The estimator $\hat{\theta}$ is an efficient GMM estimator which satisfies

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \hat{\theta}) = o_P(\|\Omega_n\|^{1/2}) \quad (3.14)$$

$$(Q'_n \Omega_n^{-1} Q_n)^{1/2} \sqrt{n}(\hat{\theta} - \theta) = (Q'_n \Omega_n^{-1} Q_n)^{-1/2} \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} Q'_n \Omega_n^{-1} \psi(X_{ig}, \theta) + o_P(1) \Rightarrow \mathcal{N}(0, I_p). \quad (3.15)$$

These are high-level assumptions that assure the Wald statistic provides valid inference under the null. We also require additional smoothness of $\mathbb{E} \psi(X_{ig}, \theta)$, $\mathbb{E} \psi(X_{ig}, \theta) \psi(X_{ig}, \theta)'$, and $\mathbb{E} \hat{\psi}(X_{ig}, \theta)$ in θ . This allows us to use the delta-method when deriving the asymptotic behaviour of the variance estimators. The moment conditions are stronger than required for $W_n \Rightarrow \chi_p^2$, however the Lindeberg conditions will be sufficient for W_n to have a limit distribution under the alternative. The assumptions as stated serve the purpose of clarifying the environment: inference settings where asymptotically normal inference is correct in large samples.

Assumption 3. We assume that

$$\begin{aligned} \|\hat{\Omega}_n - \Omega_n\| &\xrightarrow{P} 0 \\ \|\hat{Q}_n - Q_n\| &\xrightarrow{P} 0 \end{aligned}$$

where here $\|\cdot\|$ is the spectral norm. We further assume that for any r_{1n} and r_{2n} such that

$$r_{1n} \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}_{ig}(\theta) - \mathbb{E} \hat{\psi}_{ig}(\theta) \right) = O_P(1), \quad r_{1n}(\hat{\theta} - \theta) = O_P(1) \quad (3.16)$$

$$r_{2n} \left(\frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' - \Omega_n \right) = O_P(1), \quad r_{2n}(\hat{\theta} - \theta) = O_P(1) \quad (3.17)$$

we have that

$$r_{1n} \left(\hat{Q}_n - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \hat{\psi}_{ig}(\hat{\theta}) - \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}_{ig}(\theta) - \mathbb{E} \hat{\psi}_{ig}(\theta) \right) \right) = o_P(1) \quad (3.18)$$

$$r_{2n} \left(\hat{\Omega}_n - \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}_g(X_g, \hat{\theta}) \tilde{\psi}_g(X_g, \hat{\theta})' - \left(\frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' - \Omega_n \right) \right) = o_P(1). \quad (3.19)$$

In characterising the asymptotic distributions of $\hat{Q}_n$ and $\hat{\Omega}_n$, we cannot rule out that the estimation of θ impacts the asymptotic behaviour of these matrices, which take $\hat{\theta}$ as an input. This assumption will imply that the estimation of θ only enters in the first-order asymptotic behaviour of $\hat{Q}_n$ and $\hat{\Omega}_n$ through $\mathbb{E} \hat{\psi}_{ig}(\hat{\theta})$ and $\mathbb{E} \tilde{\psi}_g(X_g, \hat{\theta}) \tilde{\psi}_g(X_g, \hat{\theta})'$, which can effectively be handled using the delta-method. Furthermore, the additional noise due to estimating θ must go to zero at a faster rate than the estimation error in estimating Q_n or Ω_n if θ was known. This condition is typically satisfied when the $\hat{\psi}$ and ψ functions are sufficiently smooth in θ to satisfy a stochastic-equicontinuity condition. Differentiable functions, Lipschitz functions, and more generally Donsker classes will generally satisfy these conditions. We now discuss verification of Assumptions 2 and 3 in the context of several examples.

Example 1 (*t-statistic*). For the t-statistic with X_{ig} independent across i and g , $Q_n = 1$ and $\Omega_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} \left(\sum_{i=1}^{n_g} (X_{ig} - \mu) \right)^2 = \text{Var}(X_{ig}) = \sigma^2$. Thus, the main substantive assumptions in Assumption 2 are that $\mathbb{E} X_{ig}^4 < \infty$ and there exists some c such that $\sigma^2 \geq c > 0$. In this case we can apply a classic central limit theorem to obtain that $\frac{1}{\sigma} \sqrt{n}(\bar{X}_n - \mu) = \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} \frac{1}{\sigma} (X_{ig} - \mu) \Rightarrow \mathcal{N}(0, 1)$.

With $\mathbb{E} X_{ig}^4 < \infty$, we have that $\sqrt{n}(\hat{\sigma}_c^2 - \sigma^2) = O_P(1)$ for both the homoskedastic and cluster-robust variance estimators.³ Furthermore, we have that for $\hat{\sigma}_c^2$,

$$\begin{aligned} \mathbb{E} \hat{\sigma}_c^2 &= \sigma^2 - \frac{n_g}{n} \sigma^2 \\ \sqrt{n} \left(\hat{\sigma}_c^2 - \mathbb{E} \hat{\sigma}_c^2 - \left(\frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} X_{ig} - \mu \right)^2 - \sigma^2 \right) \right) \\ &= \sqrt{n} \left(\frac{n_g}{n} \sigma^2 - n_g (\bar{X}_n - \mu)^2 \right) \\ &\xrightarrow{P} 0 \end{aligned}$$

with a similar argument showing that $\hat{\sigma}_h^2$ also satisfies (3.19). Note that in the case of the t-test assumptions involving Q_n and $\hat{Q}_n$ are trivially satisfied as Q_n is known with $Q_n = 1$.

³Note that in the case of the t-test, the heteroskedastic-robust and homoskedastic variance estimator are the same, so the natural comparison is between homoskedastic and cluster-robust variance estimators.

Example 2 (*Efficient Smooth-GMM*). If $\psi(X_{ig}, \theta)$ is twice-continuously differentiable in θ , $\mathbb{E} \frac{\partial}{\partial \theta} \psi(X_{ig}, \theta)$ has full-rank, $\mathbb{E} \|\psi(X_{ig}, \theta)\|^2 < \infty$, and the data are i.i.d., the efficient-GMM estimator $\hat{\theta}$ will generally satisfy (3.15) with

$$Q_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \frac{\partial}{\partial \theta} \psi(X_{ig}, \theta) = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \dot{\psi}_{ig}(\theta) \quad (3.20)$$

$$\Omega_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}(X_g, \theta) \tilde{\psi}(X_g, \theta)' \quad (3.21)$$

For verifying (3.15), standard conditions involve assuming Θ is compact, Ω_n is of full rank with minimum eigenvalue bounded away from zero, Q_n is full rank, and $\sup_{i,g} \sup_{\theta \in \Theta} \dot{\psi}_{ig}(\theta) \leq C$ for some $C < \infty$. See Hansen and Lee (2019) for a full list of conditions for validity of Wald tests in a GMM setting with clustering, as well as Newey and McFadden (1994), Theorem 3.2 for a discussion of the i.i.d. case when $G = n$ and $n_g = 1$. In the smooth moment case, (3.14) will be satisfied as the first-order condition will typically hold exactly, and therefore $\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \hat{\theta}) = 0$.

The main assumptions required here that are not required for the asymptotic validity of the Wald test statistic are the existence of the second moments for the derivative matrix and fourth moments for the moment conditions; in other words, we will assume that there exists an $\epsilon > 0$ such that

$$\begin{aligned} \sup_{i,g} \sup_{\theta \in \Theta} \mathbb{E} \|\psi(X_{ig}, \theta)\|^{2r+\epsilon} &< \infty \\ \sup_{i,g} \sup_{\theta \in \Theta} \mathbb{E} \left\| \frac{\partial}{\partial \theta} \psi(X_{ig}, \theta) \right\|^{r+\epsilon} &< \infty \end{aligned}$$

where $r \geq 2$ is as in Assumption 1.

In the case of smooth GMM settings when $\psi(X_{ig}, \theta)$ is twice-differentiable in θ , then Assumption 3 will be satisfied. We can simplify the statement of Assumption 3 to specialize to the Efficient Smooth-GMM case.

Assumption 3^{gmm} (*Smoothness assumption for GMM*). Let there exist a neighbourhood N_θ of θ such that $\psi(X_{ig}, \theta)$ is twice-continuously differentiable on N_θ for all X_{ig} , and let $\psi_l(X_{ig}, \theta)$ denote the l^{th} element of the function $\psi_l(X_{ig}, \theta)$. We assume that there exists a C such that

$$\begin{aligned} \sup_{i,g,l,m} \sup_{b \in N_\theta} \mathbb{E} \left\| \left(\frac{\partial^2}{\partial b \partial b'} \psi_l(X_{ig}, b) \right) \right\|^{r+\epsilon} &\leq C < \infty \\ \sup_{i,g} \sup_{b \in N_\theta} \mathbb{E} \left\| \left(\frac{\partial}{\partial b} \psi(X_{ig}, b) \right)' \psi(X_{ig}, b) \right\|^{r+\epsilon} &\leq C < \infty \end{aligned}$$

for all $b \in N_\theta$.

Under this condition, Assumption 3 will be satisfied.

Lemma 1. Under Assumption 3^{gmm} , we have that

$$\hat{Q}_n - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \mathbb{E} \hat{\psi}_{ig}(\hat{\theta}) - \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}_{ig}(\theta) - Q_n \right) = o_P(\|\hat{\theta} - \theta\|) \quad (3.22)$$

$$\hat{\Omega}_n - \frac{1}{n} \sum_{g=1}^G \mathbb{E} \tilde{\psi}_g(X_g, \hat{\theta}) \tilde{\psi}(X_g, \hat{\theta})' - \left(\frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \theta) \tilde{\psi}(X_g, \theta)' - \Omega_n \right) = o_P(\|\hat{\theta} - \theta\|) \quad (3.23)$$

An easy way to verify the second part of Assumption 3^{gmm} is to check that $\sup_{i,g} \sup_{b \in N_\theta} \mathbb{E} \left\| \frac{\partial}{\partial b} \psi(X_{ig}, b) \right\|^{2r+\epsilon} < \infty$. Assumption 3^{gmm} can then be used to prove Lemma 1 by providing a Lipchitz-type bound on the empirical process. In the context of Assumption 3, we can choose the rates r_{1n}, r_{2n} to be of order $\frac{\sqrt{n}}{\sqrt{\ell' V_n \ell}}$.

Example 3 (*Cluster-Robust Regression*). Consider the standard linear regression model:

$$y_{ig} = x'_{ig} \theta + \varepsilon_{ig} \quad (3.24)$$

Here, $\psi(X_{ig}, \theta) = x_{ig}(y_{ig} - x'_{ig} \theta) = x_{ig} \varepsilon_{ig}$. In this case, we have

$$Q_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E} x_{ig} x'_{ig} \quad (3.25)$$

$$\Omega_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} \left[\left(\sum_{i=1}^{n_g} x_{ig} \varepsilon_{ig} \right) \left(\sum_{i=1}^{n_g} x_{ig} \varepsilon_{ig} \right)' \right]. \quad (3.26)$$

The moment conditions will be satisfied if we require that for some $\epsilon > 0$, $\mathbb{E} \|x_{ig}\|^{2r+\epsilon} < \infty$ and $\mathbb{E} y_{ig}^{2r+\epsilon} < \infty$. Standard assumptions on the design imply that $\lambda_{min}(\Omega_n)$ is bounded away from zero and Q_n is full rank. Since (3.14) is satisfied exactly for the least-squares estimator $\hat{\theta}$, (3.15) will hold.

For Assumption 3 in the regression setting, (3.18) holds trivially as it does not depend on θ . For $\hat{\Omega}_n$, the setup is similar to the t-statistic, however it is perhaps more straightforward to verify Assumption 3^{gmm} : if $\sup_{i,g} \mathbb{E} \|x_{ig}\|^{2r+\epsilon} < \infty$, in which case Assumption 3^{gmm} is satisfied. We have already assumed bounded $2r + \epsilon$ moments to verify Assumption 2.

Example 4 (*Quantile Regression*). Consider the quantile regression model:

$$y_{ig} = x'_{ig} \theta(\tau) + \varepsilon_{ig}(\tau). \quad (3.27)$$

We will assume in this example that $G = n$ and $n_g = 1$, with i.i.d. (y_{ig}, x_{ig}) . In this setting, $\psi(X_{ig}, \theta(\tau)) = x_{ig} \left(\mathbb{1}_{[y_{ig} - x'_{ig} \theta(\tau) < 0]} - \tau \right)$, and

$$\dot{\psi}_{ig}(\theta(\tau)) = \mathbb{E} x_{ig} x'_{ig} f_\tau(\varepsilon_{ig}(\tau) | x_{ig}) \quad (3.28)$$

where $f_\tau(\cdot|x_{ig})$ is the conditional density of $\varepsilon_{ig}(\tau)$ given x_{ig} . The expressions for Ω_n and Q_n are then

$$\begin{aligned} Q_n &= \dot{\psi}_{ig}(\theta(\tau)) \\ &= \mathbb{E} x_{ig} x'_{ig} f_\tau(0|x_{ig}) \\ \Omega_n &= \mathbb{E} x_{ig} x'_{ig} \left(\mathbb{1}_{[y_{ig} - x'_{ig} \theta(\tau) < 0]} - \tau \right)^2, \end{aligned}$$

When $\sup_{ig} \sup_u f_\tau(u|x_{ig}) \leq C < \infty$ and $\mathbb{E} \|x_{ig}\|^{4+\epsilon} < \infty$ then (3.15) is satisfied, and thus Assumption 2 holds.

Verification of Assumption 3 requires careful examination of the estimation of

$$Q_n = \mathbb{E} x_{ig} x'_{ig} f_\tau(0|x_{ig}) \quad (3.29)$$

which involves the unknown conditional density f_τ . A standard estimator is a kernel estimator. Let $K(u)$ be a symmetric function of bounded variation such that $\int_{\mathbb{R}} uK(u)du = 0$ and $\int_{\mathbb{R}} u^2K(u)du = 1$. A kernel estimator of Q_n is given by

$$\hat{Q}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K\left(\frac{y_{ig} - x'_{ig} \hat{\theta}_\tau}{h_n}\right) \quad (3.30)$$

where h_n is a bandwidth. Thus, we have that, in the notation we have used thus far, $\hat{\Psi}_{ig}(\hat{\theta}_\tau) = x_{ig} x'_{ig} h_n^{-1} K((y_{ig} - x'_{ig} \hat{\theta}_\tau)/h_n)$. Assumption 3 is then satisfied with the following assumptions on K and the density of the error $\varepsilon_i(\tau)$:

Assumption 3^{reg} (*Conditional density estimation for quantile regression*).

1. The kernel function K is symmetric, of bounded variation, and normalized such that:

$$\int_{\mathbb{R}} uK(u)du = 0, \quad \int_{\mathbb{R}} u^2K(u)du = 1 \quad (3.31)$$

2. There exist functions $G_j(x_{ig})$ such that for all x_{ig} , $G_j(x_{ig}) \geq |f^{(j)}(u|x_{ig})|$, uniformly in u , $j \in \{0, 1, 2\}$. Furthermore, G_j also satisfy, for some $\delta_j > 0$, $\mathbb{E}(G_0(x_{ig})\|x_{ig}\|^{4+\delta_0}) < \infty$, $\mathbb{E}(G_1(x_{ig})\|x_{ig}\|^{2+\delta_1}) < \infty$, and $\mathbb{E}(G_2(x_{ig})\|x_{ig}\|^2) < \infty$.
3. $h_n \rightarrow 0$ and $\frac{\log n}{\sqrt{nh_n}} \rightarrow 0$.

The first assumption is satisfied by all kernel functions used in practice, such as the Gaussian, Epanechnikov, Uniform, Biweight, and Triweight kernels. The assumption on bounded variation implies that the function can only rise and fall finitely many times.

The assumption on the integrability of envelope functions G_j for the density and its derivatives is quite similar to assumptions used in Kato (2012) in proving asymptotic normality of the variance estimator when using the uniform kernel. Bounding the density and the first two derivatives is standard in the literature on kernel density estimation, and in the regression context due to the conditional nature of the density we must impose additional restrictions on the regressors to ensure integrability of the effective envelope functions that are used in the bounds.

This bandwidth condition will allow the rate-optimal bandwidth, $h_n \propto n^{-1/5}$. It is slightly stronger than the bandwidth condition in Powell (1991), $n^2 h_n \rightarrow \infty$. In the estimation of $\mathbb{E} x_{ig} x'_{ig} f(0|x_{ig})$, we can

decompose our kernel estimator into three components:

$$\hat{Q}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) \quad (3.32)$$

$$= \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) \quad (3.33)$$

$$+ \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) \quad (3.34)$$

$$- \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) \right) \quad (3.35)$$

$$+ \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right). \quad (3.36)$$

Considering the estimator this way, (3.35) will be asymptotically negligible at rate $o_P(1/\sqrt{nh_n})$, and this kind of result typically makes use of empirical process methods, and that is how we proceed here in proving Lemma 2 below. For (3.36), the delta-method applies. Finally, concerning (3.33) we have to assess whether additional centring is needed. The kernel density estimator requires recentring, since although (3.33) will be consistent for Q_n , its expectation differs from Q_n and thus we need to incorporate the bias term for asymptotic normality. Then, standard results on asymptotic normality of kernel density estimators give that, for any constant matrix A ,

$$\sqrt{nh_n} \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^n \text{tr} \left(A \left(x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) \right) \right) \right) \Rightarrow \mathcal{N}(0, V(A)) \quad (3.37)$$

for $V(A) = \mathbb{E}[(x'_{ig} A x_{ig})^2 f(0|x_{ig}) R_K]$ and $R_K = \int K(u)^2 du$ is the roughness of the kernel $K(\cdot)$. Assumption $\mathcal{I}^{qreg}$ implies that this is the only component of the asymptotic distribution of $\hat{Q}_n$ which is relevant for the first-order theory:

Lemma 2. *Under Assumption $\mathcal{I}^{qreg}$, we have that*

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) \quad (3.38)$$

$$- \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) \right) = o_P(1/\sqrt{nh_n}) \quad (3.39)$$

$$\mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) = o_P(1/\sqrt{nh_n}) \quad (3.40)$$

Thus, the quantile regression model satisfies Assumption 3, where we can observe that the important rates to consider are $r_{1n} = \sqrt{nh_n}$ and $r_{2n} = \sqrt{n}$.

We are now ready to state our general result on the asymptotic distribution of W_n under fixed alternatives. First, so that the statement of theorem is focused on the key aspects, we define a sequence of constants B_n

and Ξ_n . Recall that $V_n = (Q_n' \Omega_n^{-1} Q_n)^{-1}$. Define:

$$J_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \begin{pmatrix} c_n' \frac{\partial}{\partial \theta_1} \mathbb{E} \hat{\psi}(X_{ig}, \theta) b_n \\ \vdots \\ c_n' \frac{\partial}{\partial \theta_p} \mathbb{E} \hat{\psi}(X_{ig}, \theta) b_n \end{pmatrix} \quad (3.41)$$

$$S_n = \frac{1}{n} \sum_{g=1}^G \begin{pmatrix} c_n' \frac{\partial}{\partial \theta_1} \mathbb{E} \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' c_n \\ \vdots \\ c_n' \frac{\partial}{\partial \theta_p} \mathbb{E} \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' c_n \end{pmatrix} \quad (3.42)$$

$$a_n := 2 \frac{\Delta \ell}{\ell' V_n \ell} + J_n - S_n \quad (3.43)$$

$$b_n := \frac{\Delta}{\ell' V_n \ell} 2 V_n \ell \quad (3.44)$$

$$c_n := \frac{\Delta}{\ell' V_n \ell} \Omega_n^{-1} Q_n V_n \ell. \quad (3.45)$$

Then, we can define:

$$\tilde{Q}_n := \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}(X_{ig}, \theta) \quad (3.46)$$

$$B_n := \frac{\Delta^2}{\ell' (\mathbb{E} \tilde{Q}_n' \Omega_n^{-1} \mathbb{E} \tilde{Q}_n)^{-1} \ell} - \xi_n^2 \quad (3.47)$$

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a_n' (Q_n' \Omega_n^{-1} Q_n)^{-1} Q_n' \Omega_n^{-1} \psi(X_{ig}, \theta) \\ \sum_{i=1}^{n_g} c_n' (\hat{\psi}(X_{ig}, \theta) - \mathbb{E} \hat{\psi}(X_{ig}, \theta)) b_n \\ -c_n' (\tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' - \mathbb{E} \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)') c_n \end{pmatrix} \quad (3.48)$$

$$\Xi_n := \frac{1}{n^2} \sum_{g=1}^G \mathbb{E} (\mathbf{1}' Y_{gn})^2 \quad (3.49)$$

Notice that $n\sqrt{\Xi_n} \rightarrow \infty$, as we require the number of clusters G to be increasing, and $\Xi_n = 0$ whenever $\Delta = 0$.

Theorem 1. *Assume that Assumptions 1-3 are satisfied. Suppose that $\mathbf{1}' Y_{gn}$ is nondegenerate (implying $\Xi_n > 0$). Then, for the previously defined sequences Ξ_n , B_n , we have that*

$$\frac{\frac{1}{n} W_n - \xi_n^2 - B_n}{\sqrt{\Xi_n}} \Rightarrow \mathcal{N}(0, 1) \quad (3.50)$$

Remark 1. In Hall (2000) variance estimators are considered which are potentially inconsistent under the null hypothesis, and it is shown that the power of such tests is dominated by tests constructed with variance estimators which are consistent under the null and alternative. We could embed that result in ours by allowing for B_n to include the difference $\xi_n^2 - \xi_0^2$, where ξ_0^2 is the probability limit of a Wald test statistic using a variance estimator which is only consistent under the null, such that $\frac{1}{n} W_n - \xi_0^2 \xrightarrow{P} 0$. In modern practise variance estimators are used which are consistent under the null and alternative, and thus we focus on the case where the only source of bias comes from the estimation of Q_n .

Remark 2. One could also obtain the theorem by providing sufficient conditions for joint asymptotic nor-

mality of the components of the vector (3.48). The approach here is slightly simpler, as we directly consider an asymptotic expansion of the form in (3.4), and then provide conditions under which this linear process is asymptotically normal. The intuition from the simple example of the t-test carries through in any case: the behaviour of Wald test statistics under fixed alternatives is determined by the joint asymptotic behaviour of the point estimator with the estimators for the components of the asymptotic variance.

When standard plug-in estimators for Q_n are available in GMM settings, then there will be no bias term B_n . This follows from Assumption 3, as this assumption implies the plug-in estimator is asymptotically equivalent to the infeasible estimator using the true θ , hence asymptotically unbiased. In the context of quantile regression, however, B_n will depend on the choice of bandwidth and kernel for the variance estimator. In general, however, different choices of the estimators for Q_n and Ω_n will lead to difference sequences B_n and Ξ_n . This highlights an important feature of these asymptotics: under fixed alternatives, the choice of variance estimator impacts the limit distribution of the test statistic.

We now discuss, and derive expressions for, B_n and Ξ_n in the context of our examples.

Example 1, continued (*t*-test). Focusing on the cluster-robust test statistic W_n^c , $\ell = 1$ and $V_n = \sigma^2$, so we have that $J_n = 0$, $S_n = 0$, $a_n = \frac{2\Delta}{\sigma^2}$, $c_n = \frac{\Delta}{\sigma^2}$, ignoring b_n because $Q_n = 1$. Therefore,

$$B_n = 0$$

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} \frac{2\Delta}{\sigma^2} (X_{ig} - \mu) \\ 0 \\ -\frac{\Delta^2}{\sigma^4} \left(\left(\sum_{i=1}^{n_g} (X_{ig} - \mu) \right)^2 - n_g \sigma^2 \right) \end{pmatrix}$$

Observe that in this case $\mathbf{1}'Y_{gn}$ is nondegenerate as long as X_{ig} is not a Binary random variable such that $(X_{ig} - \mu)^2$ is constant almost surely.⁴ Therefore, recalling that $\xi = \Delta/\sigma$, γ is the skewness of X_{ig} and κ is the kurtosis, we obtain that when n_g is constant for all n ,

$$\Xi_n = \frac{1}{n} [\xi^2 ((\kappa - 1)\xi^2 - 4\gamma\xi + 4 + 2(n_g - 1)\xi^2)]$$

It is also useful in this simple case to consider the general form of Ξ_n when the clusters are heterogenous. In this case

$$\mathbb{E}(\mathbf{1}'Y_{gn})^2 = 4\xi^2 n_g + \xi^4 n_g (\kappa + 2n_g - 3) - 4\xi^3 \gamma n_g$$

and therefore we have that, rearranging in a similar way to the case of fixed n_g ,

$$\Xi_n = \frac{1}{\left(\sum_{g=1}^G n_g \right)^2} \sum_{g=1}^G 4\xi^2 n_g + \xi^4 n_g (\kappa + 2n_g - 3) - 4\xi^3 \gamma n_g \quad (3.51)$$

$$= \frac{1}{n} \left[\xi^2 \left((\kappa - 1)\xi^2 - 4\gamma\xi + 4 + 2\xi^2 \frac{\sum_{g=1}^G n_g (n_g - 1)}{n} \right) \right] \quad (3.52)$$

where $\sum_{g=1}^G n_g (n_g - 1)$ is the number of additional terms we are allowing to be non-zero in the covariance matrix of the vector of observations when using the cluster-robust variance estimator, corresponding to

⁴Finding a limit distribution is still feasible in this case, but this is outside the scope of this paper; see Bentkus et al. (2007).

covariances which are 0 give that the data are i.i.d. We can re-write this again in terms of the variance of the cluster sizes. If we let $\bar{n} = \frac{1}{G} \sum_{g=1}^G n_g$ and $\nu_c^2 = \frac{1}{G} \sum_{g=1}^G (n_g - \bar{n})^2$

$$\sum_{g=1}^G n_g(n_g - 1) = \sum_{g=1}^G n_g^2 - G\bar{n} = G\nu_c^2 + G\bar{n}(\bar{n} - 1). \quad (3.53)$$

Thus, Ξ_n is increasing in both the average cluster size $\bar{n}$ and in the heterogeneity of cluster sizes (measured by ν_c^2) for a given sample size.

Example 2, continued (*Efficient Smooth-GMM*). There is not much simplification to be found from the general case.

$$J_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \begin{pmatrix} c'_n \mathbb{E} \frac{\partial}{\partial \theta \partial \theta_1} \psi(X_{ig}, \theta) b_n \\ \vdots \\ c'_n \mathbb{E} \frac{\partial}{\partial \theta \partial \theta_p} \psi(X_{ig}, \theta) b_n \end{pmatrix} \quad (3.54)$$

$$S_n = \frac{1}{n} \sum_{g=1}^G \begin{pmatrix} \sum_{l=1}^q 2c_{nl}^2 \mathbb{E} \frac{\partial}{\partial \theta_1} \tilde{\psi}_{gl}(X_g, \theta) + 2 \sum_{l < m} c_{nm} c_{nl} \mathbb{E} \left[\tilde{\psi}_{gl}(X_g, \theta) \frac{\partial}{\partial \theta_1} \psi_{gm}(X_g, \theta) \right] \\ \vdots \\ \sum_{l=1}^q 2c_{nl}^2 \mathbb{E} \frac{\partial}{\partial \theta_p} \tilde{\psi}_{gl}(X_g, \theta) + 2 \sum_{l < m} c_{nm} c_{nl} \mathbb{E} \left[\tilde{\psi}_{gl}(X_g, \theta) \frac{\partial}{\partial \theta_p} \psi_{gm}(X_g, \theta) \right] \end{pmatrix} \quad (3.55)$$

$$a_n := -2 \frac{\Delta \ell}{\ell' V_n \ell} + J_n - S_n \quad (3.56)$$

$$b_n := \frac{\Delta}{\ell' V_n \ell} 2V_n \ell \quad (3.57)$$

$$c_n := \frac{\Delta}{\ell' V_n \ell} \Omega_n^{-1} Q_n V_n \ell. \quad (3.58)$$

Example 3, continued (*Cluster-Robust Regression*). The linear setting, with plug-in estimators, implies $B_n = 0$. We also can simplify many of the expressions making up the components of Ξ_n . Note that in this case since Q_n is invertible, $V_n = Q_n^{-1} \Omega_n Q_n^{-1}$. Furthermore, Q_n and $\hat{Q}_n$ do not depend on θ or $\hat{\theta}$. Therefore, $J_n = 0$. We also have a simplification for S_n if the model is correctly specified, i.e. $\mathbb{E}[\varepsilon_g | X_g] = 0$, where X_g is the matrix with rows X'_{ig} for cluster g and let ε_g and y_g be the vector of errors and y_{ig} respectively. Then we have that $\tilde{\psi}_g(X_g, \theta) = X'_g(y_g - X'_g \theta)$. Computing S_n then gives

$$S_n = \frac{1}{n} \sum_{g=1}^G \mathbb{E} \frac{\partial}{\partial \theta} (c'_n X'_g (y_g - X'_g \theta))^2 \quad (3.59)$$

$$= -\frac{2}{n} \sum_{g=1}^G \mathbb{E} c'_n X_g \varepsilon_g X'_g X_g \quad (3.60)$$

$$= 0 \quad (3.61)$$

We then have that

$$a_n := 2 \frac{\Delta \ell}{\ell' V_n \ell} \quad (3.62)$$

$$b_n := \frac{\Delta}{\ell' V_n \ell} 2V_n \ell \quad (3.63)$$

$$c_n := \frac{\Delta}{\ell' V_n \ell} Q_n^{-1} \ell. \quad (3.64)$$

Constructing Y_{gn} gives

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a'_n x_{ig} \varepsilon_{ig} \\ \sum_{i=1}^{n_g} b'_n x_{ig} x'_{ig} c_n - \sum_{i=1}^{n_g} \mathbb{E}(b_n x_{ig} x'_{ig} c_n) \\ (\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig})^2 - \mathbb{E}(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig})^2 \end{pmatrix}. \quad (3.65)$$

The result in Theorem 1 then specialises to:

$$\frac{\frac{(\ell' \hat{\theta} - \vartheta_0)^2}{\ell' \hat{V}_n \ell} - \xi_n^2}{\sqrt{\Xi_n}} \Rightarrow \mathcal{N}(0, 1) \quad (3.66)$$

We will compute Ξ_n now in the case of correctly specified cluster dependence, and lack of cluster dependence. First, suppose that there is no cluster dependence, i.e. $\Omega_n = \mathbb{E} x_{ig} x'_{ig} \varepsilon_{ig}^2$, however a cluster-robust variance estimator is still used. Inspection of Y_{gn} tells us that the main difference is in the third element of Y_{gn}

$$\check{Y}_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a'_n x_{ig} \varepsilon_{ig} \\ \sum_{i=1}^{n_g} (b'_n x_{ig} \varepsilon_{ig})^2 - \sum_{i=1}^{n_g} \mathbb{E}(b'_n x_{ig} \varepsilon_{ig})^2 \\ \sum_{i=1}^{n_g} b'_n x_{ig} x'_{ig} c_n - \sum_{i=1}^{n_g} \mathbb{E}(b_n x_{ig} x'_{ig} c_n) \end{pmatrix}. \quad (3.67)$$

Under this assumption that the observations are i.i.d., i.e. no cluster-dependence, we have that under the null hypothesis, $\ell' \theta = \vartheta_0$,

$$\frac{n(\ell' \hat{\theta} - \vartheta_0)^2}{\ell' \hat{Q}_n^{-1} \hat{\Omega}_n \hat{Q}_n^{-1} \ell} \Rightarrow \chi_1^2 \quad (3.68)$$

$$\frac{n(\ell' \hat{\theta} - \vartheta_0)^2}{\ell' \hat{Q}_n^{-1} \check{\Omega}_n \hat{Q}_n^{-1} \ell} \Rightarrow \chi_1^2. \quad (3.69)$$

where $\check{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \hat{\varepsilon}_{ig}^2$ is the standard heteroskedastic-robust variance estimator. Thus, both statistics provide for valid inference under the null hypothesis, and have the same first-order asymptotic behaviour in that case. Let $\check{\Xi}_n$ be the normalising sequence when we use $\check{\Omega}_n$ in place of $\hat{\Omega}_n$. Comparing Ξ_n to $\check{\Xi}_n$ gives us an insight into differences in the asymptotic behaviour between the two procedures.

Proposition 1. *In the linear regression model, suppose a test of $H_0 : \ell' \theta = \vartheta_0$ can be conducted using both a cluster robust variance estimator $\hat{\Omega}_n$ and the standard heteroskedasticity-robust variance estimator $\check{\Omega}_n$:*

$$\hat{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} x_{ig} \hat{\varepsilon}_{ig} \right) \left(\sum_{i=1}^{n_g} x_{ig} \hat{\varepsilon}_{ig} \right)' \quad (3.70)$$

$$\check{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \hat{\varepsilon}_{ig}^2 \quad (3.71)$$

Then, under any fixed alternative $\vartheta \neq \vartheta_0$,

$$\frac{\frac{(\ell'\hat{\theta}-\vartheta_0)^2}{\ell'Q_n^{-1}\Omega_n Q_n^{-1}\ell} - \xi_n^2}{\sqrt{\Xi_n}} \Rightarrow \mathcal{N}(0,1) \quad (3.72)$$

$$\frac{\frac{(\ell'\hat{\theta}-\vartheta_0)^2}{\ell'Q_n^{-1}\Omega_n Q_n^{-1}\ell} - \xi_n^2}{\sqrt{\tilde{\Xi}_n}} \Rightarrow \mathcal{N}(0,1) \quad (3.73)$$

with $\Xi_n \geq \tilde{\Xi}_n$, with equality if and only if $\text{Var}(x_{ig}\varepsilon_{ig}) = 0$.

The difference $\Xi_n - \tilde{\Xi}_n$ can be directly compute as in the case of the t-statistic, and the difference is

$$\Xi_n - \tilde{\Xi}_n = \sum_{i \neq j} \mathbb{E}(c'_n x_{ig} \varepsilon_{ig})^2 \mathbb{E}(c'_n x_{jg} \varepsilon_{jg})^2 > 0 \quad (3.74)$$

which is analogous to the t-test. If in $x_{ig} = 1$ and the ε_{ig} are i.i.d. with variance σ^2 , then this simplifies to $2(n_g - 1)\xi^2$ as in the case of the t-test. We have not pinned down the sequence of significance levels α_n , and thus we cannot say that in every case $\Xi_n \geq \tilde{\Xi}_n$ implies that the cluster-robust test will be less powerful, however some exact results in Section 4 and numerical results in Section 5 indicate that this will indeed be the case in a few simple settings.

Example 4, continued (Quantile Regression). In the case of quantile regression we also obtain significant simplification, this time because the Q_n matrix is estimated at a much slower rate than θ and Ω_n . The reason for this is that Q_n is estimated at the $\sqrt{n h_n}$ rate, as is typical for kernel density estimators, while $\sqrt{n}(\hat{\theta} - \theta)$, $\sqrt{n}(\hat{\Omega}_n - \Omega_n) = O_P(1)$. This is also an example of when the bias term B_n is necessary to characterise the asymptotic theory. The expressions for b_n and c_n in this case are

$$b_n = \frac{2\Delta V_n \ell}{\ell' V_n \ell} \quad (3.75)$$

$$c_n = \frac{\Delta Q_n^{-1} \ell}{\ell' V_n \ell}. \quad (3.76)$$

Here, the bias term is, recalling that our setup for this example is i.i.d.,

$$B_n = \frac{\Delta^2}{\ell' (\mathbb{E} \tilde{Q}_n \Omega^{-1} \mathbb{E} \tilde{Q}_n)^{-1} \ell} - \xi_n^2 \quad (3.77)$$

$$= \frac{\Delta^2}{\ell' \left(Q_n + \mathbb{E} \left(x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) \right) \right)^{-1} \Omega_n \left(Q_n + \mathbb{E} \left(x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) \right) \right)^{-1} \ell} - \xi_n^2. \quad (3.78)$$

Since the bias is tending to zero and the bias term is differentiable, asymptotically we can write that

$$B_n = b'_n \mathbb{E} \left(x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) \right) c_n + o(\sqrt{n h_n}) \quad (3.79)$$

$$= \frac{1}{2} b'_n \mathbb{E}(x_{ig} x'_{ig} f''(0|x_{ig}) h_n^2) c_n + o(\sqrt{n h_n}) \quad (3.80)$$

$$(3.81)$$

Since the estimator of $\hat{\theta}$ and $\hat{\Omega}$ will converge at the parametric rate, we can also simplify our expression for Ξ_n , only the variance of $\hat{Q}_n$ is relevant:

Proposition 2. *Recall that R_K is the roughness of the kernel K , and $f(\cdot|x_{ig})$ and $f''(\cdot|x_{ig})$ are the conditional density of the error and the second derivative respectively. Then for the quantile regression model, we have that*

$$\frac{\sqrt{nh_n} \left(\frac{1}{n} W_n - \xi_n^2 - B_n \right)}{\sqrt{\mathbb{E} \left[(b'_n x_{ig} x'_{ig} c_n)^2 f(0|x_{ig}) R_K \right]}} \Rightarrow \mathcal{N}(0, 1). \quad (3.82)$$

A direct implication of this result is that the deviations of the test statistic from the non-centrality parameter in large samples are driven by the nonparametric estimator of the variance component Q_n . This is quite different from the local asymptotic theory which treats all estimators of Q_n as equivalent, as long as they are consistent. Here, the choice of kernel and bandwidth h_n play a role in the asymptotic behaviour of the test statistic.

4 Power and Sample Size Calculations

We can use the central limit-theorem result in Theorem 1 to approximate power, just as we did in Section 2. Consider the rejection probability of the standard Wald test: letting $q_{1-\alpha,p}$ be the $1 - \alpha$ -quantile of the χ_p^2 -distribution, we have that the power of a level- α test is given by:

$$P(W_n > q_{1-\alpha,p}) = P\left(\frac{\frac{1}{n} W_n - \xi_n^2 - B_n}{\sqrt{\Xi_n}} > \frac{q_{1-\alpha,p}}{n\sqrt{\Xi_n}} - \frac{\xi_n^2 + B_n}{\sqrt{\Xi_n}}\right). \quad (4.1)$$

When using local power asymptotics to compute power, practitioners generally choose an effect size Δ then use the approximation

$$P(W_n > q_{1-\alpha,p}) \approx 1 - F_{\chi_p^2(n\xi_n^2)}(q_{1-\alpha,1}). \quad (4.2)$$

The fixed alternative asymptotics imply another approximation can be used. The sample size required to achieve power $1 - \beta$ against alternative Δ at size α can be found as the solution n^* to

$$1 - \beta = 1 - \Phi\left(\frac{q_{1-\alpha,p}}{n^*\sqrt{\Xi_{n^*}}} - \frac{\xi_{n^*}^2 + B_{n^*}}{\sqrt{\Xi_{n^*}}}\right) \quad (4.3)$$

where we have approximated 4.1 with $1 - \Phi\left(\frac{q_{1-\alpha,p}}{n\sqrt{\Xi_n}} - \frac{\xi_n^2 + B_n}{\sqrt{\Xi_n}}\right)$. One could in practise produce n_1^* and n_2^* for two variance estimators for a variety of assumptions on the cluster dependence. This would produce an estimate of the efficiency loss due to robustness of the variance estimator in different scenarios. In Section 5 we use this expression to obtain sample-size calculations for the t-test when the data are i.i.d. $\mathcal{N}(\mu, \sigma^2)$, in which case $B_n = 0$, ξ_n^2 no longer depends on n , and thus the relevant terms to consider will be Ξ_n when comparing the cluster-robust t-statistic to the standard homoskedastic t-statistic.

Our formal results imply that we need to consider a sequence of critical values that diverges to ∞ , suggesting that such an approximation will be most beneficial when α is small. Our numerical results suggest that even for moderately small values of α , e.g. 0.05 and 0.01, the approximation provides an improvement over the local-power calculation.

To obtain a formal power approximation, we need to send $\alpha \rightarrow 0$. In this setup, we choose the critical

values as

$$C_n = n \left(\xi_n^2 + B_n - z_{1-\beta} \sqrt{\Xi_n} \right) \quad (4.4)$$

where again $z_{1-\beta}$ is the $1 - \beta$ -quantile of a normal distribution. Rearranging the expression, we see that:

$$P(W_n > C_n) = P \left(\frac{\frac{1}{n}W_n - \xi_n^2 - B_n}{\sqrt{\Xi_n}} > -z_{1-\beta} \right) \quad (4.5)$$

Thus, we have that $P(W_n > C_n) \rightarrow 1 - \beta$. With this formulation we are explicitly choosing a sequence of critical values such that $\alpha \rightarrow 0$. Notice that if we use (4.3) for sample-size calculations, as we do in Section 5, Ξ_n plays a role in reducing the power, or rather increasing the sample size required to achieve a particular power. Thus, results such as Proposition 1 can be translated into results on power.

We now return to our discussion of Bahadur asymptotic relative efficiency and asymptotic power calculations. To make such a connection, we must make a connection with Type-I error and power in our present setup. We can in fact provide a slight generalisation to Proposition 2 in Bahadur (1960b):

Proposition 3. *Suppose that for a sequence of tests the power $P(W_n > C_n)$ converges to a constant $1 - \beta \in (0, 1)$. Let $\alpha_n := 1 - F_{0,n}(C_n)$ denote the Type-I error for the test. Then, it must be that*

$$\frac{\log \alpha_n}{\log(1 - F_{0,n}(W_n))} \rightarrow 1. \quad (4.6)$$

Furthermore, suppose we have two tests, $W_n^{(1)}$ and $W_n^{(2)}$ with CDFs under the null hypothesis $F_{n,0}^{(1)}$ and $F_{n,0}^{(2)}$. Then, for any two sequences $C_n^{(1)}$, $C_n^{(2)}$ such that $P(W_n^{(i)} > C_n^{(i)}) \rightarrow 1 - \beta$ for each i , we have that the Bahadur asymptotic relative efficiency is given by

$$\lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^{(1)}(C_n^{(1)}))}{\log(1 - F_{n,0}^{(2)}(C_n^{(2)}))} \quad (4.7)$$

This result is also similar to Lemma 1 and the subsequent Corollary in the Appendix of Groeneboom and Oosterhoff (1981). The main difference between the statement of Proposition 3 and Proposition 2 in Bahadur (1960b) is that there is no assumption here that the correct normalising sequence for $\log \alpha_n$ is $\frac{1}{n}$, however modifying that proof to accommodate this is trivial. The proposition is straightforward, but we did not find its statement in the literature. The second part follows from the first immediately as the Bahadur asymptotic relative efficiency is defined as

$$\lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^{(1)}(W_n^{(1)}))}{\log(1 - F_{n,0}^{(2)}(W_n^{(2)}))} \quad (4.8)$$

if it exists. Rearranging and normalising by $\log \alpha_n^{(i)} = \log(1 - F_{n,0}^{(i)}(C_n^{(i)}))$, then using (4.6) leads to the result. Proposition 3 gives a recipe for finding the right sequence of tests so that the p-values under a particular alternative and Type-I error go to zero at the same rate.

1. For a test statistic W_n that can be used to test $H_0 : \mu = \mu_0$, choose an alternative $\mu \neq \mu_0$.
2. Find a sequence of critical values C_n such that under this alternative $P(W_n > C_n) \rightarrow 1 - \beta \in (0, 1)$.
3. The Type-I error rate using this sequence of critical values converges to 0 at the same rate as the p-values under the alternative $\mu \neq \mu_0$.

One difference to point out between the Bahadur approach and our power calculations is that the choice of $1 - \beta$ is irrelevant for finding the rate at which $\alpha_n \rightarrow 0$: the p-values do not depend on the critical values, and therefore only the particular alternative is relevant for the Bahadur comparison. In general, Bahadur asymptotic relative efficiency calculations are done with the choice $C_n^B = n\xi_n^2$. For our general setup, $C_n/C_n^B \rightarrow 1$. This in turn implies that any choice of C_n of the form in (4.4) leads to the same convergence rate of α_n to 0. This implies that Bahadur asymptotic relative efficiency comparisons do not depend on the limiting power of the sequence of tests; this is well known and can be discerned from Propositions 1 and 2 from Appendix 1 in Bahadur (1960b). The non-dependence of the Bahadur comparison on power simplifies the comparison of tests in some sense, however it implies that sample-size calculations using Bahadur slopes will not be as effective when targeting particular power.

For the simple example using the t-test, we can directly compute the rate at which size tends to 0 for the choices of critical values. Proposition 3 implies that we can set $z_{1-\beta} = 0$, or equivalently set $\beta = 1/2$ to find the rate at which size tends to 0. For this part, we assume that the X_{ig} are i.i.d. $\mathcal{N}(\mu, 1)$. Doing so for W_n^h and W_n^c leads to

$$-\frac{1}{n} \log(1 - F_{n,0}^h(n\xi^2)) \rightarrow \frac{1}{2} \log(1 + \xi^2) \quad (4.9)$$

$$-\frac{1}{n} \log(1 - F_{n,0}^c(n\xi^2)) \rightarrow \frac{1}{2n_g} \log\left(1 + \frac{\xi^2}{n_g}\right). \quad (4.10)$$

The expression for W_n^h is classic and can be found in e.g. Serfling (2009). The expression for W_n^c can be derived by considering the $Y_g = \sum_{i=1}^{n_g} X_{ig}$ as the observations and applying the same arguments used to derive (4.9) as $G \rightarrow \infty$. The additional n_g terms come from $n = Gn_g$ when all clusters are of the same size.

This comparison gives the convergence rate of Type-I error to zero under a sequence of tests which lead to power converging to 1/2. If $n_g = 1$ the two expressions are the same. The ratio

$$\frac{n_g \log(1 + \xi^2)}{\log(1 + \xi^2/n_g)} \quad (4.11)$$

is maximised at $\xi^2 = 0$, tends to n_g as $\xi^2 \rightarrow \infty$, and it can be shown that this ratio monotonically decreasing in ξ^2 and thus is always greater than 1, uniformly in ξ^2 , as long as $n_g > 1$.

A simple way of identifying the preference for the homoskedastic test using W_n^h can be seen based on the asymptotic power analysis. Based on the above calculations for the Type-I error, we have that

$$\frac{\log(1 - F_{n,0}^h(n\xi^2))}{\log(1 - F_{n,0}^c(nn_g((\xi^2 + 1)^{n_g} - 1)))} \rightarrow 1 \quad (4.12)$$

To see this, observe that

$$\frac{1}{2n_g} \log\left(1 + \frac{n_g((\xi^2 + 1)^{n_g} - 1)}{n_g}\right) = \frac{1}{2} \log(1 + \xi^2)$$

Thus, if we specify the tests as rejecting when $W_n^h > n\xi^2$ and $W_n^c > nn_g((\xi^2 + 1)^{n_g} - 1)$, we can compare the tests with respect to power, as Type-I error converges to zero for each test at the same rate. We have that

$$P(W_n^h > n\xi^2) \rightarrow 1/2 \quad (4.13)$$

and thus need only consider W_n^c . Computing the power function using our approximation, we have

$$P(W_n^c > nn_g((\xi^2 + 1)^{n_g} - 1)) = P\left(\sqrt{n}\left(\frac{1}{n}W_n^c - \xi^2\right) > \sqrt{n}(n_g(\xi^2 + 1)^{n_g} - (\xi^2 + 1))\right) \rightarrow 0.$$

We can generalise this in the following proposition:

Proposition 4. *Under the assumption that the data X_{ig} are i.i.d. $\mathcal{N}(\mu, 1)$, with $\xi^2 = (\mu - \mu_0)^2$, for any sequences of critical values C_n^h, C_n^c such that $P(W_n^h > C_n^h) \rightarrow 1 - \beta \in (0, 1)$ and*

$$\frac{\log(1 - F_{n,0}^h(C_n^h))}{\log(1 - F_{n,0}^c(C_n^c))} \rightarrow 1 \quad (4.14)$$

we have that

$$P(W_n^c > C_n^c) \rightarrow 0. \quad (4.15)$$

To see that the proposition is true, consider first that determining the convergence rate of Type-I error to zero does not depend on the asymptotic power $1 - \beta$, by Proposition 3. Thus, for any C_n^h , we can use the choice $C_n^c = nn_g((\xi^2 + 1)^{n_g} - 1)$. For this sequence, $P(W_n^c > C_n^c) \rightarrow 0$, and thus the proposition is proved.

This analysis might suggest that considering the power of the tests as size tends to zero does not necessarily add new insights in the comparison of tests beyond that of the Bahadur approach. In general, the standard goal of relative efficiency is different from power analysis: try to understand the necessary sample size to attain a particular size and power combination for a given alternative. Bahadur asymptotic relative efficiency approximates the necessary sample size to achieve power $1 - \beta$ at alternative $\mu \neq \mu_0$ by sending size to 0. Let $n_a(\alpha, \beta, \mu)$ be the sample size necessary for $P(W_n^a > C_n^a) \geq 1 - \beta$ with $1 - F_{0,n}^a(C_n^a) \leq \alpha_n$. Along the chosen sequence, $\alpha_n \rightarrow 0$. Then, Bahadur approximates the relative efficiency $n_c(\alpha_n, \beta, \mu)/n_h(\alpha_n, \beta, \mu)$ based on the fact that

$$\lim_{n \rightarrow \infty} \frac{n_c(\alpha_n, \beta, \mu)}{n_h(\alpha_n, \beta, \mu)} = \lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^h(C_n^h))}{\log(1 - F_{n,0}^c(C_n^c))} = \lim_{n \rightarrow \infty} \frac{\log(1 - F_{n,0}^h(W_n^h))}{\log(1 - F_{n,0}^c(W_n^c))}.$$

In all of these expressions, β drops out, but as discussed in Groeneboom and Oosterhoff (1981), in general computing $n_c(\alpha_n, \beta, \mu)/n_h(\alpha_n, \beta, \mu)$ will depend on β . Thus, one can understand the computation of the limiting distribution of the test statistics under a fixed alternative as providing a clarification of how to interpret Bahadur asymptotic relative efficiency. We can also consider the analysis in Kallenberg (1981), in which they considered the Bahadur deficiency, $n_c(\alpha_n, \beta, \mu) - n_h(\alpha_n, \beta, \mu)$. In that paper, Theorem 1.2 gives an expression for the power of the test in terms of the p-values. They point out that the deficiency still depends on β in general. We show in the simulation section that sample size calculations using our setup can provide more accurate results, as they allow for the resulting sample size to depend on β .

5 Numerical results

In this section we present a set of numerical results for the simple example of the one-sample t-test from Section 2 in which we compute the sample size required for a particular power. To be precise, we will use several different approximations to compute the minimum sample size required to achieve power at least

$1 - \beta$ against alternative Δ at size α . We will study the behaviour of the classical t-test statistic (up to scaling) and the cluster robust version:

$$W_h = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_h^2}, \quad \hat{\sigma}_h^2 = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} (X_{ig} - \bar{X}_n)^2$$

$$W_c = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_c^2}, \quad \hat{\sigma}_c^2 = \frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} (X_{ig} - \bar{X}_n) \right)^2.$$

We only consider the case the X_{ig} are i.i.d. $\mathcal{N}(\Delta, 1)$, with $H_0 : \mathbb{E} X_{ig} = 0$. In this case, $\xi_n^2 = \Delta^2$. Thus, $\frac{n-1}{n} W_h$ has an exact $F_{1, n-1}(n\xi^2)$ distribution. When cluster sizes are homogenous, then we also have an exact distribution for W_c , $\frac{G-1}{G} W_c \sim F_{1, G-1}(n\xi^2)$. In these cases we can compute the true required sample size up to numerical precision with standard statistical software. This also implies in our asymptotic power calculations, which we will outline below, we can use quantiles of the appropriate F distribution for the critical values.

In the case of heterogenous cluster sizes, we will parametrise the cluster sizes in the same way as in MacKinnon, Nielsen, and Webb (2023a):

$$n_g = n \frac{\exp\{\gamma g/G\}}{\sum_{j=1}^G \exp\{\gamma j/G\}}. \quad (5.1)$$

In MacKinnon et al. (2023a) they used $[n_g]$, where $[\cdot]$ takes the integer part, for their cluster sizes. We will use a Satterthwaite approximation and avoid using simulation. We do this for two reasons. The first reason is computational. If we used Monte Carlo, we would need to search over sample sizes by first finding the correct critical value under the null hypothesis, then evaluating whether power was above or below the designated threshold $1 - \beta$. Thus, the number of Monte Carlo iterations required depends on the quality of initial guesses and would have been a limiting factor in the diversity of experiments we could run. The second reason is practical. Our results are conditional on a sequence of cluster sizes, but the finite-sample properties may be sensitive to the order in which clusters arrive, and whether large clusters arrive first relative to small clusters is an arbitrary choice we would rather avoid taking a stance on.

Instead of Monte Carlo methods, we use a Satterthwaite approximation and in the end our cluster sizes and heterogeneity are parametrised by two parameters: γ from (5.1), and the average cluster size, $\bar{n} := \frac{1}{G} \sum_{g=1}^G n_g$. Full details are given in the appendix. The degree of heterogeneity in clusters sizes is dictated by γ . If $\gamma = 0$, then all clusters are the same size.

We consider two asymptotic approximations for sample size calculations in this section, and also compare with sample size calculations based on Bahadur slopes in the appendix. Bahadur is arguably less suited to a sample size calculation as the approximation is invariant to the target power. The primary approximations we consider are one based on local power, and the fixed-alternative approximation that is the primary focus in this paper.

For the local power approximation, we will use a $\chi_1^2(n\Delta^2)$ to perform power calculations. The difference between sample size calculations for W_h and W_c will then come from the differences in critical values used. For our fixed-alternative approximation, we will use (4.3), noting that in this case $B_n = 0$. Ξ_n can be found

by specialising (3.52) to the case of normally distributed observations, so that $\gamma = 0$ and $\kappa = 3$, giving

$$\Xi_n = \frac{1}{n} \left[\xi^2 \left(2\xi^2 + 4 + 2\xi^2 \frac{\sum_{g=1}^G n_g(n_g - 1)}{n} \right) \right]. \quad (5.2)$$

For our numerical results we vary the size of the tests ($\alpha \in \{0.01, 0.05\}$), the target power ($(1 - \beta) \in \{0.8, 0.9\}$), the alternative considered ($\Delta \in \{0.5, 1.0, 2.0\}$), the average cluster size ($\bar{n} \in \{20, 500, 1000\}$), and the degree of heterogeneity in cluster sizes ($\gamma \in \{0, 4, 8\}$).

For each data-generating process we fix size and compute the minimum sample size required to achieve a particular power. For Pitman, with $W_n \Rightarrow \chi_1^2(\xi^2)$, we will use the $\chi_1^2(n\xi^2)$ distribution to come up with the estimated sample size. For fixed-alternative approximation, we will use (4.3). That is, we will compute the sample size required to achieve power at least $1 - \beta$ against alternative Δ at size α . We will normalise the variance to be 1 so that $\xi^2 = \Delta^2$. The required sample size will be denoted n^* and will be computed by finding a solution to

$$1 - \beta = 1 - \Phi \left(\frac{C_\alpha}{n^* \sqrt{\Xi_{n^*}}} - \frac{\xi_{n^*}^2 + B_{n^*}}{\sqrt{\Xi_{n^*}}} \right) \quad (??) \quad (5.3)$$

where in each case we will choose C_α such that the test has size α under the null.⁵ Note that when we use Bahadur to perform this exercise, the sample size calculation does not depend on β , and instead uses

$$n_B^* = -\frac{2 \log \alpha}{c(\xi^2)}, \quad c(\xi^2) = \lim_{n \rightarrow \infty} \frac{\log \alpha_n}{n}. \quad (5.4)$$

We focus on the 1-sample t-test. We will study the behaviour of the classical t-test statistic (up to scaling) and the cluster robust version:

$$W_h = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_h^2}, \quad \hat{\sigma}_h^2 = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} (X_{ig} - \bar{X}_n)^2$$

$$W_c = \frac{n(\bar{X}_n - \mu_0)^2}{\hat{\sigma}_c^2}, \quad \hat{\sigma}_c^2 = \frac{1}{n} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} (X_{ig} - \bar{X}_n) \right)^2.$$

In all our numerical experiments, the X_{ig} are i.i.d. $\mathcal{N}(\mu, 1)$. Thus, $\frac{n-1}{n} W_h$ has an exact $F_{1, n-1}(n\xi^2)$ distribution. When cluster sizes are homogenous, then we also have an exact distribution for W_c , $\frac{G-1}{G} W_c \sim F_{1, G-1}(n\xi^2)$.

We now present some numerical results on the t-test. All results in these tables were computed without relying on Monte Carlo methods. The results are summarised in Tables 1-4. We report the sample size N_h and N_c required for the tests using W_h and W_c respectively to achieve power $1 - \beta$ at size α given average cluster size $\bar{n}$ and heterogeneity γ against the alternative Δ . We approximate N_c using N_c^{np} , N_c^p , N_c^b , and N_c^* . N_c^{np} is the ‘naïve’ Pitman approximation using the same critical value from the $F_{1, n-1}$ distribution as in the homoskedastic case. We view this as the ‘standard’ asymptotic approximation with critical value approximately equal to the χ_1^2 critical values $q_{1-\alpha, 1}$. N_c^p uses a local approximation with the same critical value from the calculation of N_c , thus either the $F_{1, G-1}$ critical values or approximation using Satterthwaite approximate degrees of freedom. N_c^b uses the Bahadur slope as in (5.4). N_c^* uses (??). The *ERR* terms correspond to the relative error in approximating N_c using N_c^{np} , N_c^p , N_c^b , and N_c^* . For sample size estimate

⁵If we did not first choose C_α and instead used $q_{1-\alpha, p}$, differences in accuracy of sample-size calculation could be confounded with differences in the validity of the cluster-robust test statistic in finite samples.

$\hat{n}$, the relative error $ERR = \frac{\hat{n} - N_c}{N_c}$.

Δ	$\bar{n}$	γ	N_h	N_c	N_c^{np}	N_c^p	N_c^b	N_c^*	ERR_c^{np}	ERR_c^p	ERR_c^b	ERR_c^*
0.50	20	0	33.37	76.83	33.81	80.64	153.76	60.63	-0.56	0.05	1.00	-0.21
2.00	20	0	4.22	42.21	4.44	43.42	74.54	40.40	-0.89	0.03	0.77	-0.04
5.00	20	0	2.49	35.94	2.58	36.85	58.55	35.34	-0.93	0.03	0.63	-0.02
0.50	500	0	33.37	1006.40	33.81	1033.92	1738.86	971.86	-0.97	0.03	0.73	-0.03
2.00	500	0	4.22	829.47	4.44	849.39	1288.21	823.69	-0.99	0.02	0.55	-0.01
5.00	500	0	2.49	768.56	2.58	786.03	1135.12	768.71	-1.00	0.02	0.48	0.00
0.50	1000	0	33.37	1891.73	33.81	1941.01	3168.68	1846.13	-0.98	0.03	0.68	-0.02
2.00	1000	0	4.22	1606.72	4.44	1644.52	2444.72	1600.76	-1.00	0.02	0.52	-0.00
5.00	1000	0	2.49	1502.23	2.58	1535.64	2183.30	1505.14	-1.00	0.02	0.45	0.00
0.50	20	4	33.37	101.13	33.81	111.15	227.47	82.81	-0.67	0.10	1.25	-0.18
2.00	20	4	4.22	44.38	4.44	52.38	111.35	45.60	-0.90	0.18	1.51	0.03
5.00	20	4	2.49	32.72	2.58	38.92	81.99	35.56	-0.92	0.19	1.51	0.09
0.50	500	4	33.37	1012.47	33.81	1204.87	2564.13	1062.91	-0.97	0.19	1.53	0.05
2.00	500	4	4.22	720.74	4.44	833.81	1695.57	783.02	-0.99	0.16	1.35	0.09
5.00	500	4	2.49	655.39	2.58	728.73	1367.95	701.00	-1.00	0.11	1.09	0.07
0.50	1000	4	33.37	1796.49	33.81	2148.14	4561.41	1930.33	-0.98	0.20	1.54	0.07
2.00	1000	4	4.22	1381.12	4.44	1572.46	3113.42	1492.46	-1.00	0.14	1.25	0.08
5.00	1000	4	2.49	1279.41	2.58	1405.45	2544.80	1360.26	-1.00	0.10	0.99	0.06
0.50	20	8	33.37	146.56	33.81	166.14	350.93	128.35	-0.77	0.13	1.39	-0.12
2.00	20	8	4.22	51.48	4.44	79.04	184.86	66.37	-0.91	0.54	2.59	0.29
5.00	20	8	2.49	27.26	2.58	53.26	136.23	45.82	-0.91	0.95	4.00	0.68
0.50	500	8	33.37	1032.66	33.81	1786.99	4268.39	1508.78	-0.97	0.73	3.13	0.46
2.00	500	8	4.22	610.73	4.44	1027.05	2771.92	907.91	-0.99	0.68	3.54	0.49
5.00	500	8	2.49	575.51	2.58	788.99	2135.80	729.20	-1.00	0.37	2.71	0.27
0.50	1000	8	33.37	1559.36	33.81	3074.31	7599.93	2618.11	-0.98	0.97	3.87	0.68
2.00	1000	8	4.22	1187.54	4.44	1836.80	5020.20	1650.75	-1.00	0.55	3.23	0.39
5.00	1000	8	2.49	1135.45	2.58	1467.92	3870.00	1377.81	-1.00	0.29	2.41	0.21

Table 1: $\alpha = 0.05$, $1 - \beta = 0.8$.

The first observation is in comparing N_h and N_c . Clearly it uniformly requires fewer observations to achieve power at a given alternative using W_h instead of W_c , unsurprisingly since W_h is the test statistic for the uniformly most powerful unbiased test.

Now, consider the efficacy of the different approximations. Using the naïve N_c^{np} is a poor approximation of N_c , however in general for small values of the alternative, smaller cluster sizes, and low-moderate cluster heterogeneity, N_c^p provides a better approximation to N_c than N_c^* and N_c^b . N_c^b does quite poorly in general, and is dominated by N_c^p and N_c^* in our numerical exercise.

As expected, N_c^p does quite well, especially for $\gamma = 0$. In these cases, we are effectively using the approximation of the $F_{1,n-1}$ distribution with the χ_n^2 distribution. We start to observe N_c^* as a strong competitor when Δ gets larger, $\bar{n}$ gets larger, and γ gets larger. In fact, with heterogenous cluster sizes ($\gamma > 0$), N_c^* dominates almost uniformly N_c^p in terms of relative error whenever $\Delta = 5$, the lone exception occurring when $\alpha = 0.01$ and $1 - \beta = 0.9$.

6 Conclusion

In this paper we develop a first-order asymptotic theory of Wald test statistics under fixed alternatives. We motivate this discussion by mapping the asymptotic distribution to a relative efficiency measure. Our

Δ	$\bar{n}$	γ	N_h	N_c	N_c^{np}	N_c^p	N_c^b	N_c^*	ERR_c^{np}	ERR_c^p	ERR_c^b	ERR_c^*
0.50	20	0	50.06	113.93	50.35	115.72	225.62	87.31	-0.56	0.02	0.98	-0.23
2.00	20	0	6.31	56.41	6.41	56.87	103.84	52.43	-0.89	0.01	0.84	-0.07
5.00	20	0	3.46	45.89	3.49	46.36	79.26	44.14	-0.92	0.01	0.73	-0.04
0.50	500	0	50.06	1327.88	50.35	1339.19	2404.43	1247.89	-0.96	0.01	0.81	-0.06
2.00	500	0	6.31	1032.42	6.41	1044.79	1711.66	1006.80	-0.99	0.01	0.66	-0.02
5.00	500	0	3.46	931.98	3.49	944.55	1476.34	918.89	-1.00	0.01	0.58	-0.01
0.50	1000	0	50.06	2452.58	50.35	2475.45	4333.79	2335.62	-0.98	0.01	0.77	-0.05
2.00	1000	0	6.31	1978.54	6.41	2003.58	3220.88	1938.85	-1.00	0.01	0.63	-0.02
5.00	1000	0	3.46	1806.73	3.49	1831.70	2819.03	1786.50	-1.00	0.01	0.56	-0.01
0.50	20	4	50.06	159.26	50.35	164.32	338.07	127.18	-0.68	0.03	1.12	-0.20
2.00	20	4	6.31	73.53	6.41	78.00	165.33	68.85	-0.91	0.06	1.25	-0.06
5.00	20	4	3.46	52.43	3.49	57.32	122.77	52.03	-0.93	0.09	1.34	-0.01
0.50	500	4	50.06	1679.80	50.35	1794.49	3813.03	1597.37	-0.97	0.07	1.27	-0.05
2.00	500	4	6.31	1066.75	6.41	1189.44	2561.58	1096.13	-0.99	0.12	1.40	0.03
5.00	500	4	3.46	870.24	3.49	976.87	2091.64	917.66	-1.00	0.12	1.40	0.05
0.50	1000	4	50.06	2951.37	50.35	3189.31	6804.10	2869.41	-0.98	0.08	1.31	-0.03
2.00	1000	4	6.31	1956.89	6.41	2194.14	4725.94	2036.85	-1.00	0.12	1.42	0.04
5.00	1000	4	3.46	1643.70	3.49	1838.64	3905.10	1739.14	-1.00	0.12	1.38	0.06
0.50	20	8	50.06	238.90	50.35	248.41	522.39	200.58	-0.79	0.04	1.19	-0.16
2.00	20	8	6.31	114.37	6.41	125.79	279.07	110.06	-0.94	0.10	1.44	-0.04
5.00	20	8	3.46	73.70	3.49	89.58	210.73	78.97	-0.95	0.22	1.86	0.07
0.50	500	8	50.06	2578.47	50.35	2884.51	6474.67	2532.81	-0.98	0.12	1.51	-0.02
2.00	500	8	6.31	1206.51	6.41	1755.92	4406.95	1545.52	-0.99	0.46	2.65	0.28
5.00	500	8	3.46	727.99	3.49	1278.55	3573.29	1128.87	-1.00	0.76	3.91	0.55
0.50	1000	8	50.06	4381.20	50.35	5072.15	11630.93	4467.48	-0.99	0.16	1.65	0.02
2.00	1000	8	6.31	1875.31	6.41	3112.36	8119.24	2737.98	-1.00	0.66	3.33	0.46
5.00	1000	8	3.46	1356.44	3.49	2278.15	6634.95	2024.98	-1.00	0.68	3.89	0.49

Table 2: $\alpha = 0.01, 1 - \beta = 0.8$.

main finding is that this alternative asymptotic framework distinguishes between approaches to testing that more classical approaches cannot. This opens up the possibility of comparing different variance estimators with testing in mind. This contrasts with previous comparisons that have previously been made based on simulation evidence, higher-order comparisons, or finite-sample criteria. Our approach applies to a broad class of models. One conclusion of particular interest for applied researchers is that there is an asymptotic justification for assuming there is a cost for clustering at too-coarse a level.

There are also plenty of cases of interest not considered here. Our analysis could be applied to comparing commonly used heteroskedastic-robust variance estimators. We also did not pursue any high-dimensional or machine learning applications here, and it would be interesting to consider how our power analysis could provide guidance for tuning parameter choices in that setting. We could also extend the results to accomodate settings where the variance estimator is not asymptotically normal, as is the case when the 4th moment of the influence function does not exist. The theory in Bentkus et al. (2007) introduces similar results in the case of the 1-sample t-test, and similar results could be produced in a more general setting, such as in the case of linear processes driven by i.i.d. infinite-variance innovations (see, for example, Cavaliere, Georgiev, and Taylor (2016)).

Δ	$\bar{n}$	γ	N_h	N_c	N_c^{np}	N_c^p	N_c^b	N_c^*	ERR_c^{np}	ERR_c^p	ERR_c^b	ERR_c^*
0.50	20	0	50.06	113.93	50.35	115.72	225.62	87.31	-0.56	0.02	0.98	-0.23
2.00	20	0	6.31	56.41	6.41	56.87	103.84	52.43	-0.89	0.01	0.84	-0.07
5.00	20	0	3.46	45.89	3.49	46.36	79.26	44.14	-0.92	0.01	0.73	-0.04
0.50	500	0	50.06	1327.88	50.35	1339.19	2404.43	1247.89	-0.96	0.01	0.81	-0.06
2.00	500	0	6.31	1032.42	6.41	1044.79	1711.66	1006.80	-0.99	0.01	0.66	-0.02
5.00	500	0	3.46	931.98	3.49	944.55	1476.34	918.89	-1.00	0.01	0.58	-0.01
0.50	1000	0	50.06	2452.58	50.35	2475.45	4333.79	2335.62	-0.98	0.01	0.77	-0.05
2.00	1000	0	6.31	1978.54	6.41	2003.58	3220.88	1938.85	-1.00	0.01	0.63	-0.02
5.00	1000	0	3.46	1806.73	3.49	1831.70	2819.03	1786.50	-1.00	0.01	0.56	-0.01
0.50	20	4	50.06	159.26	50.35	164.32	338.07	127.18	-0.68	0.03	1.12	-0.20
2.00	20	4	6.31	73.53	6.41	78.00	165.33	68.85	-0.91	0.06	1.25	-0.06
5.00	20	4	3.46	52.43	3.49	57.32	122.77	52.03	-0.93	0.09	1.34	-0.01
0.50	500	4	50.06	1679.80	50.35	1794.49	3813.03	1597.37	-0.97	0.07	1.27	-0.05
2.00	500	4	6.31	1066.75	6.41	1189.44	2561.58	1096.13	-0.99	0.12	1.40	0.03
5.00	500	4	3.46	870.24	3.49	976.87	2091.64	917.66	-1.00	0.12	1.40	0.05
0.50	1000	4	50.06	2951.37	50.35	3189.31	6804.10	2869.41	-0.98	0.08	1.31	-0.03
2.00	1000	4	6.31	1956.89	6.41	2194.14	4725.94	2036.85	-1.00	0.12	1.42	0.04
5.00	1000	4	3.46	1643.70	3.49	1838.64	3905.10	1739.14	-1.00	0.12	1.38	0.06
0.50	20	8	50.06	238.90	50.35	248.41	522.39	200.58	-0.79	0.04	1.19	-0.16
2.00	20	8	6.31	114.37	6.41	125.79	279.07	110.06	-0.94	0.10	1.44	-0.04
5.00	20	8	3.46	73.70	3.49	89.58	210.73	78.97	-0.95	0.22	1.86	0.07
0.50	500	8	50.06	2578.47	50.35	2884.51	6474.67	2532.81	-0.98	0.12	1.51	-0.02
2.00	500	8	6.31	1206.51	6.41	1755.92	4406.95	1545.52	-0.99	0.46	2.65	0.28
5.00	500	8	3.46	727.99	3.49	1278.55	3573.29	1128.87	-1.00	0.76	3.91	0.55
0.50	1000	8	50.06	4381.20	50.35	5072.15	11630.93	4467.48	-0.99	0.16	1.65	0.02
2.00	1000	8	6.31	1875.31	6.41	3112.36	8119.24	2737.98	-1.00	0.66	3.33	0.46
5.00	1000	8	3.46	1356.44	3.49	2278.15	6634.95	2024.98	-1.00	0.68	3.89	0.49

Table 3: $\alpha = 0.05, 1 - \beta = 0.9$.

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Δ	$\bar{n}$	γ	N_h	N_c	N_c^{np}	N_c^p	N_c^b	N_c^*	ERR_c^{np}	ERR_c^p	ERR_c^b	ERR_c^*
0.50	20	0	62.87	128.67	62.75	124.67	225.62	86.07	-0.51	-0.03	0.75	-0.33
2.00	20	0	7.19	59.62	6.99	57.36	103.84	52.12	-0.88	-0.04	0.74	-0.13
5.00	20	0	3.73	47.74	3.57	46.47	79.26	43.97	-0.93	-0.03	0.66	-0.08
0.50	500	0	62.87	1396.56	62.75	1347.42	2404.43	1241.29	-0.96	-0.04	0.72	-0.11
2.00	500	0	7.19	1066.34	6.99	1045.71	1711.66	1003.66	-0.99	-0.02	0.61	-0.06
5.00	500	0	3.73	956.47	3.57	944.80	1476.34	916.72	-1.00	-0.01	0.54	-0.04
0.50	1000	0	62.87	2563.76	62.75	2484.62	4333.79	2325.00	-0.98	-0.03	0.69	-0.09
2.00	1000	0	7.19	2037.96	6.99	2004.71	3220.88	1933.44	-1.00	-0.02	0.58	-0.05
5.00	1000	0	3.73	1850.79	3.57	1832.02	2819.03	1782.66	-1.00	-0.01	0.52	-0.04
0.50	20	4	62.87	178.64	62.75	173.53	338.07	125.28	-0.65	-0.03	0.89	-0.30
2.00	20	4	7.19	79.67	6.99	78.77	165.33	68.19	-0.91	-0.01	1.08	-0.14
5.00	20	4	3.73	56.37	3.57	57.52	122.77	51.62	-0.94	0.02	1.18	-0.08
0.50	500	4	62.87	1816.25	62.75	1808.06	3813.03	1582.85	-0.97	-0.00	1.10	-0.13
2.00	500	4	7.19	1138.64	6.99	1191.30	2561.58	1088.83	-0.99	0.05	1.25	-0.04
5.00	500	4	3.73	914.74	3.57	977.39	2091.64	913.03	-1.00	0.07	1.29	-0.00
0.50	1000	4	62.87	3182.15	62.75	3205.55	6804.10	2845.16	-0.98	0.01	1.14	-0.11
2.00	1000	4	7.19	2077.83	6.99	2196.44	4725.94	2024.53	-1.00	0.06	1.27	-0.03
5.00	1000	4	3.73	1717.13	3.57	1839.26	3905.10	1731.37	-1.00	0.07	1.27	0.01
0.50	20	8	62.87	265.21	62.75	257.80	522.39	197.84	-0.76	-0.03	0.97	-0.25
2.00	20	8	7.19	125.34	6.99	126.78	279.07	108.88	-0.94	0.01	1.23	-0.13
5.00	20	8	3.73	82.49	3.57	89.87	210.73	78.12	-0.96	0.09	1.55	-0.05
0.50	500	8	62.87	2832.70	62.75	2902.65	6474.67	2505.85	-0.98	0.02	1.29	-0.12
2.00	500	8	7.19	1419.77	6.99	1759.14	4406.95	1528.73	-1.00	0.24	2.10	0.08
5.00	500	8	3.73	792.47	3.57	1279.62	3573.29	1117.29	-1.00	0.61	3.51	0.41
0.50	1000	8	62.87	4845.95	62.75	5095.12	11630.93	4420.10	-0.99	0.05	1.40	-0.09
2.00	1000	8	7.19	2242.30	6.99	3116.63	8119.24	2708.33	-1.00	0.39	2.62	0.21
5.00	1000	8	3.73	1427.88	3.57	2279.52	6634.95	2005.78	-1.00	0.60	3.65	0.40

Table 4: $\alpha = 0.01$, $1 - \beta = 0.9$.

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A Proofs of Main Results

A.1 Proof of Theorem 1

Proof. The delta-method can be used to find the asymptotic distribution of

$$\frac{(\ell'\hat{\theta} - \vartheta_0)^2}{\ell'\hat{V}_n\ell}. \quad (\text{A.1})$$

Recall $\tilde{Q}_n$ and define $\tilde{\Omega}_n$:

$$\tilde{Q}_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}(X_{ig}, \theta) \quad (\text{A.2})$$

$$\tilde{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)'. \quad (\text{A.3})$$

We first expand to isolate the variance estimator:

$$\begin{aligned} \frac{(\ell'\hat{\theta} - \vartheta_0)^2}{\ell'\hat{V}_n\ell} - \xi_n^2 - B_n &= \frac{(\ell'\hat{\theta} - \vartheta)^2}{\ell'\hat{V}_n\ell} + 2 \frac{\Delta(\ell'\hat{\theta} - \vartheta)}{\ell'\hat{V}_n\ell} + \frac{\Delta^2}{\ell'\hat{V}_n\ell} - \xi_n^2 - B_n \\ &= \frac{(\ell'\hat{\theta} - \vartheta)^2}{\ell'\hat{V}_n\ell} + 2 \frac{\Delta(\ell'\hat{\theta} - \vartheta)}{\ell'\hat{V}_n\ell} \\ &\quad + \frac{\Delta^2}{\ell'(\hat{Q}'_n \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(Q'_n \Omega_n^{-1} Q_n)^{-1} \ell} - B_n \\ &\quad + \frac{\Delta^2}{\ell'(\hat{Q}'_n \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} \\ L_{1n} &= \frac{(\ell'\hat{\theta} - \vartheta)^2}{\ell'\hat{V}_n\ell} + 2 \frac{\Delta(\ell'\hat{\theta} - \vartheta)}{\ell'\hat{V}_n\ell} \\ L_{2n} &= \frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(Q'_n \Omega_n^{-1} Q_n)^{-1} \ell} - B_n \\ L_{3n} &= \frac{\Delta^2}{\ell'(\hat{Q}'_n \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell}. \end{aligned}$$

L_{1n} is the contribution of $\hat{\theta}$, L_{2n} is the contribution of $\hat{Q}_n$, and L_{3n} is the contribution of $\hat{\Omega}_n$.

For L_{1n} , we have that

$$\sqrt{\ell'V_n\ell}\sqrt{n}L_{1n} = \frac{2\Delta}{\sqrt{\ell'V_n\ell}} \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} \ell'(Q'_n \Omega_n^{-1} Q_n)^{-1} Q'_n \Omega_n^{-1} \psi(X_{ig}, \theta) + o_P(1) \quad (\text{A.4})$$

by (3.15). Define

$$b_n := \frac{\Delta}{\ell'V_n\ell} 2V_n\ell \quad (\text{A.5})$$

$$c_n := \frac{\Delta}{\ell'V_n\ell} \Omega_n^{-1} Q_n V_n \ell. \quad (\text{A.6})$$

Now, focusing on L_{2n} , applying the delta-method gives us

$$\frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(Q'_n \Omega_n^{-1} Q_n)^{-1} \ell} - B_n = \frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} \quad (\text{A.7})$$

$$- \frac{\Delta^2}{\ell'(\mathbb{E} \tilde{Q}'_n \Omega^{-1} \mathbb{E} \tilde{Q}_n)^{-1} \ell} \quad (\text{A.8})$$

$$= \frac{\Delta^2}{(\ell' V_n \ell)^2} \text{tr}(2V_n \ell \ell' V_n Q'_n \Omega_n^{-1} (\hat{Q}_n - \mathbb{E} \tilde{Q}_n)) \quad (\text{A.9})$$

$$+ o_P(\|\hat{Q}_n - \mathbb{E} \tilde{Q}_n\|) \quad (\text{A.10})$$

$$= c'_n(\hat{Q}_n - \mathbb{E} \tilde{Q}_n) b_n + o_P(|c'_n(\hat{Q}_n - \mathbb{E} \tilde{Q}_n) b_n|) \quad (\text{A.11})$$

Note that we can rewrite

$$\hat{Q}_n - \mathbb{E} \tilde{Q}_n = \tilde{Q}_n - \mathbb{E} \tilde{Q}_n \quad (\text{A.12})$$

$$+ \mathbb{E} \hat{Q}_n - \mathbb{E} \tilde{Q}_n \quad (\text{A.13})$$

$$+ [\hat{Q}_n - \mathbb{E} \hat{Q}_n - (\tilde{Q}_n - \mathbb{E} \tilde{Q}_n)] \quad (\text{A.14})$$

By (3.18),

$$\|[\hat{Q}_n - \mathbb{E} \hat{Q}_n - (\tilde{Q}_n - \mathbb{E} \tilde{Q}_n)]\| = o_P(\|\hat{\theta} - \theta\| + \|\tilde{Q}_n - \mathbb{E} \tilde{Q}_n\|) \quad (\text{A.15})$$

Therefore, if we define

$$J_n = \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \begin{pmatrix} c'_n \frac{\partial}{\partial \theta_1} \mathbb{E} \hat{\psi}(X_{ig}, \theta) b_n \\ \vdots \\ c'_n \frac{\partial}{\partial \theta_p} \mathbb{E} \hat{\psi}(X_{ig}, \theta) b_n \end{pmatrix} \quad (\text{A.16})$$

we can write L_{2n} as

$$L_{2n} = c'_n(\tilde{Q}_n - \mathbb{E} \tilde{Q}_n) b_n + J'_n(\hat{\theta} - \theta) + o_P(|J'_n(\hat{\theta} - \theta)| + |c'_n(\tilde{Q}_n - \mathbb{E} \tilde{Q}_n) b_n|) \quad (\text{A.17})$$

Lastly, working with L_{3n} , we have

$$L_{3n} = \frac{\Delta^2}{\ell'(\hat{Q}'_n \hat{\Omega}_n^{-1} \hat{Q}_n)^{-1} \ell} - \frac{\Delta^2}{\ell'(\hat{Q}'_n \Omega_n^{-1} \hat{Q}_n)^{-1} \ell} \quad (\text{A.18})$$

$$= -\frac{\Delta^2}{(\ell' V_n \ell)^2} \text{tr}(\Omega_n^{-1} Q_n V_n \ell \ell' V_n Q'_n \Omega_n^{-1} (\hat{\Omega}_n - \Omega_n)) + o_P(\|\hat{\Omega}_n - \Omega_n\|) \quad (\text{A.19})$$

$$= -c'_n(\hat{\Omega}_n - \Omega_n) c_n + o_P(|c'_n(\hat{\Omega}_n - \Omega_n) c_n|) \quad (\text{A.20})$$

Defining

$$\tilde{\Omega}_n = \frac{1}{n} \sum_{g=1}^G \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' \quad (\text{A.21})$$

$$S_n = \frac{1}{n} \sum_{g=1}^G \begin{pmatrix} c'_n \frac{\partial}{\partial \theta_1} \mathbb{E} \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' c_n \\ \vdots \\ c'_n \frac{\partial}{\partial \theta_p} \mathbb{E} \tilde{\psi}_g(X_g, \theta) \tilde{\psi}_g(X_g, \theta)' c_n \end{pmatrix}. \quad (\text{A.22})$$

we then write

$$L_{3n} = -c'_n(\tilde{\Omega}_n - \Omega_n)c_n - S'_n(\hat{\theta} - \theta) + o_P(|S'_n(\hat{\theta} - \theta)| + |c'_n(\hat{\Omega}_n - \Omega_n)c_n|) \quad (\text{A.23})$$

Therefore, defining $a_n = 2\frac{\Delta\ell}{\ell'\hat{V}_n\ell} + J_n - S_n$, we have shown that

$$\frac{(\ell'\hat{\theta} - \vartheta_0)^2}{\ell'\hat{V}_n\ell} - \xi_n^2 - B_n = a'_n(\hat{\theta} - \theta) + \quad (\text{A.24})$$

$$+ c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n \quad (\text{A.25})$$

$$- c'_n(\tilde{\Omega}_n - \Omega_n)c_n \quad (\text{A.26})$$

$$+ o_P(|a'_n\hat{\theta} - \theta|) + o_P(|c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n|) + o_P(|c'_n\hat{\Omega}_n - \Omega_n)c_n|. \quad (\text{A.27})$$

By Assumption 3, we can ignore these o_P terms if we find a normalising sequence for the other terms: that is to say if we find a sequence Ξ_n such that

$$\Xi_n^{-1/2} \left(a'_n(\hat{\theta} - \theta) + c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n - c'_n(\tilde{\Omega}_n - \Omega_n)c_n \right) = O_P(1) \quad (\text{A.28})$$

We have that

$$\Xi_n^{-1/2} \left(\frac{(\ell'\hat{\theta} - \vartheta_0)^2}{\ell'\hat{V}_n\ell} - \xi_n^2 - B_n - \left(a'_n(\hat{\theta} - \theta) + c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n - c'_n(\tilde{\Omega}_n - \Omega_n)c_n \right) \right) = o_P(1) \quad (\text{A.29})$$

We will use Theorem 2 in Hansen and Lee (2019). Define

$$Y_{gn} = \begin{pmatrix} \sum_{i=1}^{n_g} a'_n(Q'_n\Omega_n^{-1}Q_n)^{-1}Q'_n\Omega_n^{-1}\psi(X_{ig}, \theta) \\ \sum_{i=1}^{n_g} c'_n(\hat{\psi}(X_{ig}, \theta) - \mathbb{E}\hat{\psi}(X_{ig}, \theta))b_n \\ -c'_n(\tilde{\psi}_g(X_g, \theta)\tilde{\psi}_g(X_g, \theta)' - \mathbb{E}\tilde{\psi}_g(X_g, \theta)\tilde{\psi}_g(X_g, \theta)')c_n \end{pmatrix} \quad (\text{A.30})$$

Note that $\mathbb{E}Y_{gn} = 0$, $\frac{1}{n}\sum_{g=1}^G Y_{gn} = a'_n(\hat{\theta} - \theta) + c'_n(\tilde{Q}_n - \mathbb{E}\tilde{Q}_n)b_n - c'_n(\tilde{\Omega}_n - \Omega_n)c_n$, and define $\Xi_n = \mathbb{E}\sum_{g=1}^G (\mathbf{1}'Y_{gn})^2$. The uniform integrability condition in Assumption 2 and Assumption 1 gives us that

$$\Xi_n^{-1/2} \sum_{g=1}^G \mathbf{1}'Y_{gn} \Rightarrow \mathcal{N}(0, 1) \quad (\text{A.31})$$

by Theorem 2 in Hansen and Lee (2019), with the assumption that $\mathbf{1}'Y_{gn}$ is nondegenerate for all n, g . Finally, by Assumption (3),

$$\Xi_n^{-1/2} \left(\frac{1}{n}W_n - \xi^2 - B_n - \sum_{g=1}^G Y_{gn} \right) = o_P(1). \quad (\text{A.32})$$

□

A.2 Proof of Proposition 1

Proof. Let $\tilde{\Xi}_n = \mathbb{E}(\mathbf{1}'\check{Y}_{ng})^2$, where $\check{Y}_{ng}$ is formed using $\check{\Omega}_n$ in place of $\hat{\Omega}_n$. When computing $\Xi_n - \tilde{\Xi}_n$, we obtain a great simplification that when looking at $\mathbb{E}Y_{gn}Y'_{gn} - \mathbb{E}\check{Y}_{gn}\check{Y}'_{gn}$, we note that all terms are zero,

except for the second diagonal element. We can compute this component as a function of the moments of

$$[\mathbb{E} Y_{gn} Y'_{gn}]_{2,2} - [\mathbb{E} \check{Y}_{gn} \check{Y}'_{gn}]_{2,2} = \sum_{i \neq j} \mathbb{E}(c'_n x'_{ig} \varepsilon_{ig})^2 \mathbb{E}(c'_n x_{jg} \varepsilon_{jg})^2 \geq 0 \quad (\text{A.1})$$

where we use the notation $[\cdot]_{a,b}$ to denote the (a,b) element of the relevant matrix. To see why (A.1) is true, notice that the components of $[\check{Y}_{gn}]_2$ and $[Y_{gn}]_2$ are mean-zero. However, the variance for the second component of $\check{Y}_{gn}$ is

$$\text{Var} \left(\sum_{i=1}^{n_g} (c'_n x_{ig} \varepsilon_{ig})^2 \right) = n_g \text{Var}((c'_n x_{ig} \varepsilon_{ig})^2). \quad (\text{A.2})$$

Under Assumption 1, the cluster-robust variance estimator is still consistent. However, now we must assess the variances of these variance estimators:

$$\text{Var} \left(\left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^2 \right) = \mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^4 - \left(\mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^2 \right)^2 \quad (\text{A.3})$$

$$= \mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^4 - n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2. \quad (\text{A.4})$$

We can write this fourth-moment in terms of the fourth cumulant, which we will denote $k_4(\cdot)$, and use the property that the cumulant of the sum is the sum of cumulants:

$$\mathbb{E} \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right)^4 = k_4 \left(\sum_{i=1}^{n_g} c'_n x_{ig} \varepsilon_{ig} \right) + 3n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.5})$$

$$= \sum_{i=1}^{n_g} k_4(c'_n x_{ig} \varepsilon_{ig}) + 3n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.6})$$

$$= n_g k_4(c'_n x_{ig} \varepsilon_{ig}) + 3n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.7})$$

Thus, note that:

$$[\mathbb{E} Y_{gn} Y'_{gn}]_{3,3} - [\mathbb{E} \check{Y}_{gn} \check{Y}'_{gn}]_{3,3} = \text{Var} \left(\left(\sum_{i=1}^{n_g} (c'_n x_{ig} \varepsilon_{ig}) \right)^2 \right) - \text{Var} \left(\sum_{i=1}^{n_g} (c'_n x_{ig} \varepsilon_{ig})^2 \right) \quad (\text{A.8})$$

$$= n_g k_4(c'_n x_{ig} \varepsilon_{ig}) + 2n_g^2 \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.9})$$

$$- n_g k_4(c'_n x_{ig} \varepsilon_{ig}) - 2n_g \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.10})$$

$$= 2n_g(n_g - 1) \text{Var}(c'_n x_{ig} \varepsilon_{ig})^2 \quad (\text{A.11})$$

$$= 2 \sum_{i \neq j} \mathbb{E}(c'_n x_{ig} \varepsilon_{ig})^2 \mathbb{E}(c'_n x_{jg} \varepsilon_{jg})^2 > 0 \quad (\text{A.12})$$

This expression is completely analogous to the case with the sample mean. $\square$

A.3 Proof of Proposition 2

Proof. As a result of Lemma 2, and since we know that $\hat{\theta} - \theta = O_P(1/\sqrt{n})$ and $\hat{\Omega}_n - \Omega_n = O_P(1/\sqrt{n})$. Standard results on kernel density estimation lead to the expression for B_n and that for any matrix A we have that:

$$\sqrt{nh_n} \left(\frac{1}{n} \sum_{i=1}^n b'_n x_{ig} x'_{ig} c_n \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}}{h_n} \right) - \xi^2 - B_n \right) \Rightarrow \mathcal{N} \left(0, \mathbb{E}(b'_n x_{ig} x'_{ig} c_n)^2 f(0|x_{ig}) R_K \right) \quad (\text{A.1})$$

□

B Proofs of Lemmas

B.1 Proof of Lemma 1

A Taylor expansion gives us that, for $\hat{\theta} \in N_\theta$.

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \frac{\partial}{\partial \theta} \psi(X_{ig}, \hat{\theta}) - \dot{\psi}(\hat{\theta}) - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \frac{\partial}{\partial \theta} \psi(X_{ig}, \theta) - \dot{\psi}(\theta) \quad (\text{B.1})$$

$$= \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \sum_{k=1}^p \left(\frac{\partial^2}{\partial \theta \partial \theta'} \psi_k(X_{ig}, \bar{\theta}) - \mathbb{E} \frac{\partial^2}{\partial \theta \partial \theta'} \psi_k(X_{ig}, \bar{\theta}) \right)' (\hat{\theta} - \theta) \quad (\text{B.2})$$

$$= o_P(1) O_P(1/\sqrt{n}) = o_P(1/\sqrt{n}) \quad (\text{B.3})$$

by the assumption of bounded derivatives and the fact that $\sqrt{n}(\hat{\theta} - \theta) = O_P(1)$. Since $P(\hat{\theta} \in N_\theta) \rightarrow 1$, for any neighborhood N_θ , the result follows. Similarly, for

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \hat{\theta}) \psi(X_{ig}, \hat{\theta})' - \Psi(\hat{\theta}) - \frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \theta) \psi(X_{ig}, \theta)' - \Psi(\theta) \quad (\text{B.4})$$

$$= \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \bar{\theta})' \frac{\partial}{\partial \theta} \psi(X_{ig}, \bar{\theta})' - \mathbb{E} \left(\psi(X_{ig}, \bar{\theta})' \frac{\partial}{\partial \theta} \psi(X_{ig}, \bar{\theta})' \right) \right) (\hat{\theta} - \theta) \quad (\text{B.5})$$

$$= o_P(1/\sqrt{n}) \quad (\text{B.6})$$

B.2 Proof of Lemma 2

Proof. Boundedness of the kernel function K and the existence of integrable functions G_j which uniformly bound the error density and the derivatives implies that

$$\mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) = O_P(1/\sqrt{n}) \quad (\text{B.1})$$

via the delta-method. Now, we focus on:

$$\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\hat{\varepsilon}_{ig}(\tau)}{h_n} \right) \quad (\text{B.2})$$

$$- \left(\frac{1}{n} \sum_{g=1}^G \sum_{i=1}^{n_g} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) - \mathbb{E} x_{ig} x'_{ig} \frac{1}{h_n} K \left(\frac{\varepsilon_{ig}(\tau)}{h_n} \right) \right) \quad (\text{B.3})$$

$$= \frac{1}{\sqrt{n}} (\mathbb{G}_n(x x' \hat{f}(\hat{\theta})) - \mathbb{G}_n(x x' \hat{f}(\theta))) \quad (\text{B.4})$$

using the common notational shorthand $\mathbb{G}_n \psi(\theta) = \frac{1}{\sqrt{n}} \sum_{g=1}^G \sum_{i=1}^{n_g} \psi(X_{ig}, \theta) - \mathbb{E} \psi(X_{ig}, \theta)$. An implication of Assumption 3^{reg} is that there exist functions K_1, K_2 such that K_{ig} is non-negative, non-decreasing, and $K = K_1 - K_2$. Furthermore, $|K|_v = |K_1|_v + |K_2|_v$, so we have a simple form of the total-variation norm. Using arguments similar to those found in Einmahl and Mason (2000), we have that, for $t, s \in \mathbb{R}^p$, letting $\delta_t = t - \theta, \delta_s = s - \theta$,

$$\begin{aligned} K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) &= K_1 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K_1 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \\ &\quad - \left(K_2 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K_2 \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right) \\ &= \int_{\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}}^{\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n}} dK_1(x) - \int_{\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}}^{\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n}} dK_2(x) \end{aligned}$$

This implies, via the triangle inequality,

$$\left| K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right| \leq \int \left| \mathbb{1}_{\left[\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}, \frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right]}(x) \right| d(K_1(x) + K_2(x)) \quad (\text{B.5})$$

Thus, using (B.5), we can use Hölder's inequality to bound the mean-squared difference:

$$\mathbb{E} \left[\left(K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right)^2 \middle| x_{ig} \right] \quad (\text{B.6})$$

$$\leq \int \mathbb{E} \left| \mathbb{1}_{\left[\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n}, \frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right]}(x) \right| d(K_1(x) + K_2(x)) |K|_v \quad (\text{B.7})$$

$$= \int \left| \int_{x'_{ig} \delta_s + h_n x}^{x'_{ig} \delta_t + h_n x} f(\varepsilon | x_{ig}) d\varepsilon \right| d(K_1(x) + K_2(x)) |K|_v \quad (\text{B.8})$$

$$\leq \|f(\cdot | x_{ig})\|_\infty |K|_v^2 \|x_{ig}\|_2 \|t - s\|_2 \quad (\text{B.9})$$

Now, by Assumption 3^{reg}:

$$\mathbb{E} \left[\left(K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_t}{h_n} \right) - K \left(\frac{\varepsilon_{ig} - x'_{ig} \delta_s}{h_n} \right) \right)^2 \right] = O(\|t - s\|_2) \quad (\text{B.10})$$

Putting this all together, for any $\delta > 0$, let $N_{\delta/\sqrt{n}}(\theta)$ be a $\delta/\sqrt{n}$ neighborhood of θ . Then, we have that

for any $\epsilon > 0$,

$$P \left(\sup_{b \in N_{\delta/\sqrt{n}}(\theta)} \left| \mathbb{G}_n(xx'(\hat{f}(b) - \hat{f}(\theta))) \right| > h_n^{1/2}\epsilon \right) \quad (\text{B.11})$$

$$\leq \frac{1}{\epsilon h_n^{1/2}} \mathbb{E} \left(\sup_{b \in N_{\delta/\sqrt{n}}(\theta)} \left| \mathbb{G}_n(xx'(\hat{f}(b) - \hat{f}(\theta))) \right| \right) \quad (\text{B.12})$$

We now need a slight extension of a VC-class result from Giné and Nickl (2016):

Lemma 3. *Let $\mathcal{K} = \{(\varepsilon, x) \mapsto K\left(\frac{\varepsilon - x't}{h}\right) : t \in \mathbb{R}^p, h > 0\}$. Then $\mathcal{K}$ is of VC-type.*

The arguments are the same as in Giné and Nickl (2016), with the finite-dimensional vector space having dimension $p + 2$, so we omit the proof.

We are now ready to use the maximal inequality of Chernozhukov, Chetverikov, and Kato (2014):

$$h_n^{-1/2} \mathbb{E} \left(\sup_{b \in N_{\delta/\sqrt{n}}(\theta)} \left| \mathbb{G}_n(\psi_b - \psi_\theta) \right| \right) = O \left(\sqrt{\frac{\log n}{h_n n^{1/2}}} \right)$$

where in the notation of Corollary 5.1 of Chernozhukov et al. (2014), we can choose $\sigma^2 = O(1/\sqrt{n})$ by (B.10). This means that when $\frac{\log n}{h_n \sqrt{n}} = o(1)$, we obtain the desired result:

$$\frac{1}{\sqrt{n}} \mathbb{G}_n(xx'(\hat{f}(b) - \hat{f}(\theta))) = o_P(1/\sqrt{nh_n}) \quad (\text{B.13})$$

□

C Numerical Exercise Details

As stated in the main body of the text, we use the parametrisation of cluster sizes from MacKinnon et al. (2023a):

$$n_g = \left\lceil N \frac{\exp\{\gamma g/G\}}{\sum_{j=1}^G \exp\{\gamma j/G\}} \right\rceil \quad (\text{C.1})$$

When computing Ξ_n , this leads to a simplification:

$$\Xi_n = \frac{2\xi^2}{n} \left(2 + \frac{1}{2}\xi^2 \frac{(1 + e^\gamma)\bar{n}\gamma}{e^\gamma - 1} \right). \quad (\text{C.2})$$

To obtain the Satterthwaite approximation used in the paper, we use the approach in Hansen (2022). Let $J = \frac{1}{n}\mathbf{1}\mathbf{1}'$, the projection matrix for obtaining the sample mean, and let C be the $n \times G$ matrix where observation i is in cluster g if $C_{ig} = 1$, otherwise $C_{ig} = 0$. Then, if we define $\Lambda = \frac{1}{n}(I - J)CC'(I - J)$, we need to compute

$$a = \sqrt{\text{tr}(\Lambda)} \quad (\text{C.3})$$

$$= \sqrt{1 - \frac{\sum_{g=1}^G n_g^2}{n^2}} \quad (\text{C.4})$$

$$K = \frac{\text{tr}(\Lambda)^2}{\text{tr}(\Lambda\Lambda)} \quad (\text{C.5})$$

$$= \frac{\left(1 - \frac{1}{n^2} \sum_{g=1}^G n_g^2\right)^2}{\frac{1}{n^2} \left(\sum_{g=1}^G n_g^2 - \frac{2}{n} \sum_{g=1}^G n_g^3 + \frac{1}{n^2} \left(\sum_{g=1}^G n_g^2 \right)^2 \right)} \quad (\text{C.6})$$

Using the parametrisation of the cluster heterogeneity, and using the fact that $\bar{n} = \frac{n}{G}$, we end up with

$$a = \sqrt{1 - \frac{(1 - e^{\bar{n}\gamma/n})(1 + e^\gamma)}{(1 - e^\gamma)(1 + e^{\bar{n}\gamma/n})}} \quad (\text{C.7})$$

$$K = a^2 - 2 \frac{(1 - e^{\bar{n}\gamma/n})^2 (e^{2\gamma} + e^\gamma + 1)}{(e^{2\bar{n}\gamma/n} + e^{\bar{n}\gamma/n} + 1)((1 - e^\gamma)^2)} + a^4 \quad (\text{C.8})$$

We then use the same Satterthwaite approximation as in Hansen (2022),

$$W_c \stackrel{\sim}{\sim} \frac{1}{a^2} F_{1,K}(n\Delta^2) \quad (\text{C.9})$$